Econometrics

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Introduction

Econometrics studies how statistical models can be used to extract quantitative information from observed data. Although its applications are motivated by economic questions, much of its mathematical foundation belongs to probability, statistics, and linear algebra. In particular, the central object of this course is the regression model

$$Y = X\beta + u,$$

where Y is the response variable, X contains the explanatory variables, β is an unknown parameter vector, and u represents the part of the response that is not explained by the observed regressors.

The starting point is ordinary least squares (OLS). Given a sample, OLS chooses the parameter values that minimize the sum of squared residuals. The computational definition is simple, but the statistical behavior of the resulting estimator depends strongly on the assumptions imposed on the error term and the explanatory variables. Questions of unbiasedness, consistency, variance, standard errors, efficiency, and sampling distributions therefore form the theoretical core of the subject.

The basic linear regression model provides a useful reference point. Under suitable exogeneity and homoskedasticity assumptions, OLS is unbiased and, by the Gauss–Markov theorem, is the best linear unbiased estimator. With an additional normality assumption, exact finite-sample inference can be developed using t- and F-distributions. When sample sizes become large, laws of large numbers and central limit theorems provide a more general asymptotic framework, leading to consistency and asymptotic normality of OLS estimators.

Much of econometric analysis concerns what happens when the assumptions of the basic model are weakened. Heteroskedasticity changes the variance of the estimator and motivates robust standard errors and weighted least squares. Time series data introduce temporal dependence, trends, stationarity, and serial correlation. Misspecified functional forms, measurement error, omitted variables, and sample selection can further distort estimation and inference. Most importantly, when explanatory variables are correlated with the error term, OLS generally becomes inconsistent. Instrumental variables and two-stage least squares then provide an alternative framework for estimation.

Accordingly, these notes are organized around a sequence of statistical questions rather than around particular economic applications:

> How is the regression model specified?
>
> How are its parameters estimated?
>
> What are the statistical properties of the estimators?
>
> How can uncertainty be quantified and hypotheses tested?
>
> What fails when the classical assumptions are violated?
>
> Which alternative estimators remain valid in those settings?

Economic examples provide interpretations of these questions, but the main emphasis is on the mathematical structure of regression, estimation, and statistical inference.

Chapter 1

The Linear Regression Model and OLS

1.1 Econometrics and Economic Data

Econometrics uses statistical methods to analyze economic and other observational data. Its central objective is to describe relationships between variables, estimate unknown parameters, quantify uncertainty, and, under appropriate assumptions, study causal effects. A typical econometric model has the form

$$y = f(x_1, x_2, \ldots, x_k, u),$$

where y is the dependent variable, $x_1, \ldots, x_k$ are observed explanatory variables, and u collects all unobserved factors that affect y.
Unlike data generated by a controlled experiment, economic data are usually *nonexperimental*. Individuals, firms, prices, wages, education levels, and policy variables are observed rather than assigned by the researcher. Consequently, statistical association alone does not automatically imply causality.

Example 1.1.1 (Education and Wage)**.** Suppose the objective is to study how education is related to wage. A simple model is

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + u.$$

The error term u may contain ability, family background, occupation, work experience, motivation, and many other factors that are not explicitly included in the regression.
The coefficient β_1 therefore has a causal interpretation only under additional assumptions concerning the relationship between education and these omitted factors.

1.2 Ceteris Paribus and Causal Interpretation

A central idea in regression analysis is the *ceteris paribus* interpretation: the effect of one explanatory variable is studied while holding other relevant factors fixed.
Suppose

$$y = \beta_0 + \beta_1 x + u.$$

The coefficient β_1 measures the change in the systematic part of y associated with a one-unit change in x:

$$\Delta y = \beta_1 \Delta x,$$

provided that the unobserved component u is held fixed.
The main difficulty in observational data is that changes in x may also be associated with changes in u. Regression assumptions are therefore required to formalize when a ceteris paribus interpretation is justified.

1.3 The Simple Linear Regression Model

Model 1.3.1 (Simple Linear Regression Model)**.** The simple linear regression model is

$$y = \beta_0 + \beta_1 x + u.$$

Here,

- y is the dependent variable;
- x is the explanatory variable;
- β_0 is the intercept;

- β_1 is the slope coefficient;
- u is the error term.

The model is *linear in parameters*. The explanatory variable itself does not have to enter the model linearly. For example,

$$y = \beta_0 + \beta_1 x^2 + u$$

is still a linear regression model because it is linear in β_0 and β_1.
The error term is not necessarily a measurement error. More generally, it represents all determinants of y that are omitted from the specified regression model.

1.4 The Zero Conditional Mean Assumption

The fundamental assumption of the simple regression model is that the average value of the error term does not depend on the explanatory variable.

Assumption 1.4.1 (Zero Conditional Mean)**.**

$$\mathbb{E}(u \mid x) = 0.$$

This assumption implies

$$\mathbb{E}(u) = \mathbb{E}[\mathbb{E}(u \mid x)] = 0$$

and

$$\mathbb{E}(xu) = \mathbb{E}[x\mathbb{E}(u \mid x)] = 0.$$

Therefore,

$$\mathrm{Cov}(x, u) = \mathbb{E}(xu) - \mathbb{E}(x)\mathbb{E}(u) = 0.$$

Thus zero conditional mean implies that x and u are uncorrelated. The converse is generally not true: zero covariance is weaker than zero conditional mean.
The assumption is often interpreted as an *exogeneity* condition. If x is correlated with u, then x is called *endogenous*; otherwise it is called *exogenous* in the regression.

Example 1.4.2 (Education and Unobserved Ability)**.** Consider

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + u.$$

If unobserved ability is part of u and people with greater ability also tend to obtain more education, then education and u are positively correlated. In this case,

$$\mathbb{E}(u \mid \text{educ}) \neq 0,$$

so the zero conditional mean assumption fails.
This is an example of an endogeneity problem and illustrates why a regression coefficient need not automatically represent a causal effect.

1.5 The Population Regression Function

Under the zero conditional mean assumption,

$$\mathbb{E}(y \mid x) = \mathbb{E}(\beta_0 + \beta_1 x + u \mid x) = \beta_0 + \beta_1 x.$$

Therefore the conditional mean of y given x is linear:

$$\boxed{\mathbb{E}(y \mid x) = \beta_0 + \beta_1 x}.$$

This function is called the *population regression function.*
The regression model can therefore be decomposed into

$$y = \underbrace{\mathbb{E}(y \mid x)}_{\text{systematic part}} + \underbrace{u}_{\text{unsystematic part}}.$$

The slope coefficient can also be written in terms of population moments. Since

$$\operatorname{Cov}(x, y) = \operatorname{Cov}(x, \beta_0 + \beta_1 x + u) = \beta_1 \operatorname{Var}(x)$$

under $\operatorname{Cov}(x, u) = 0$, it follows that

$$\boxed{\beta_1 = \frac{\operatorname{Cov}(x, y)}{\operatorname{Var}(x)}}.$$

Then

$$\boxed{\beta_0 = \mathbb{E}(y) - \beta_1 \mathbb{E}(x)}.$$

These formulas describe the population coefficients. In practice the population moments are unknown and must be estimated from sample data.

1.6 Ordinary Least Squares Estimation

Suppose a random sample

$$(x_1, y_1), \ldots, (x_n, y_n)$$

is observed. For any proposed coefficients b_0 and b_1, define the residual for observation i by

$$e_i = y_i - b_0 - b_1 x_i.$$

Ordinary least squares chooses the coefficients that minimize the sum of squared residuals.

Definition 1.6.1 (Ordinary Least Squares)**.** The OLS estimators are

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{b_0, b_1} \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2.$$

The first-order conditions are

$$\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

and

$$\sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

Define

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

Then the first normal equation gives

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Substituting this expression into the second normal equation yields

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}.$$

Hence

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}.$$

The fitted regression line is therefore

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Example 1.6.2 (Wage and Education)**.** Suppose the fitted regression is

$$\widehat{\text{wage}} = 3.20 + 1.15\,\text{educ}.$$

The estimated slope indicates that one additional year of education is associated with an increase of 1.15 units in predicted wage.
This statement describes an estimated association. A causal interpretation additionally requires the relevant exogeneity assumptions.

1.7 Fitted Values and Residuals

For each observation, the fitted value is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i,$$

and the OLS residual is

$$\hat{u}_i = y_i - \hat{y}_i.$$

Therefore

$$y_i = \hat{y}_i + \hat{u}_i.$$

The OLS first-order conditions imply several useful algebraic properties.

Proposition 1.7.1 (Algebraic Properties of OLS Residuals)**.** For a simple linear regression with an intercept,

$$\sum_{i=1}^{n} \hat{u}_i = 0,$$

$$\sum_{i=1}^{n} x_i \hat{u}_i = 0,$$

and

$$\sum_{i=1}^{n} \hat{y}_i \hat{u}_i = 0.$$

Moreover,

$$\bar{\hat{y}} = \bar{y}.$$

Since

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

the fitted regression line passes through the sample point $(\bar{x}, \bar{y})$.

1.8 Measures of Variation and Goodness of Fit

The total variation in the dependent variable is measured by the total sum of squares

$$\text{SST} = \sum_{i=1}^{n}(y_i - \bar{y})^2.$$

The variation explained by the regression is

$$\text{SSE} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2,$$

while the unexplained variation is

$$\text{SSR} = \sum_{i=1}^{n}\hat{u}_i^2.$$

For an OLS regression containing an intercept,

$$\boxed{\text{SST} = \text{SSE} + \text{SSR}}.$$

The coefficient of determination is defined as

$$\boxed{R^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}}.$$

Thus R^2 measures the fraction of sample variation in y that is explained by the fitted regression.
In a simple regression with an intercept,

$$R^2 = \text{Corr}(x, y)^2.$$

A large R^2 does not by itself imply that the regression has a causal interpretation, nor does a small R^2 imply that the estimated coefficient is unimportant.

1.9 Functional Forms

Linear regression refers to linearity in the unknown parameters, not necessarily linearity in the observed variables. Transformations of x and y are therefore frequently used.

1.9.1 Level–Level Model

The model

$$y = \beta_0 + \beta_1 x + u$$

implies approximately

$$\Delta y = \beta_1 \Delta x.$$

Thus a one-unit increase in x is associated with a β_1-unit change in y.

1.9.2 Log–Level Model

Consider

$$\log(y) = \beta_0 + \beta_1 x + u.$$

For a small change in x,

$$100\frac{\Delta y}{y} \approx 100\beta_1 \Delta x.$$

Thus a one-unit increase in x is associated with an approximate $100\beta_1\%$ change in y.

Example 1.9.1 (Log Wage and Education)**.** A commonly used wage equation is

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + u.$$

If $\hat{\beta}_1 = 0.08$, one additional year of education is associated with approximately an 8% increase in wage.

1.9.3 Level–Log Model

Consider

$$y = \beta_0 + \beta_1 \log(x) + u.$$

Since

$$\Delta \log(x) \approx \frac{\Delta x}{x},$$

a 1% increase in x is associated approximately with a $0.01\beta_1$-unit change in y.

1.9.4 Log–Log Model

Consider

$$\log(y) = \beta_0 + \beta_1 \log(x) + u.$$

Then

$$\frac{\Delta y}{y} \approx \beta_1 \frac{\Delta x}{x}.$$

Therefore β_1 is an *elasticity*: a 1% increase in x is associated approximately with a β_1% change in y.

Example 1.9.2 (Price Elasticity of Demand)**.** Suppose

$$\log(Q) = \beta_0 + \beta_1 \log(P) + u,$$

where Q denotes quantity demanded and P denotes price. Then β_1 measures the price elasticity of demand.
For example, if $\hat{\beta}_1 = -1.4$, a 1% increase in price is associated with approximately a 1.4% decrease in quantity demanded.

1.10 Unbiasedness of the OLS Estimators

Consider the sample regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

Let

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Then

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n} (x_i - \bar{x}) u_i}{S_{xx}}.$$

Under the zero conditional mean assumption,

$$\mathbb{E}(u_i \mid x_1, \ldots, x_n) = 0.$$

Therefore,

$$\mathbb{E}(\hat{\beta}_1 \mid X) = \beta_1,$$

where $X = (x_1, \ldots, x_n)$. Hence

$$\boxed{\mathbb{E}(\hat{\beta}_1) = \beta_1}.$$

Similarly,

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

implies

$$\boxed{\mathbb{E}(\hat{\beta}_0) = \beta_0}.$$

Thus OLS is unbiased under the appropriate exogeneity assumption.

1.11 Homoskedasticity and the Variance of OLS

To derive the conditional variances of the OLS estimators, impose an additional assumption.

Assumption 1.11.1 (Homoskedasticity)**.**

$$\mathrm{Var}(u \mid x) = \sigma^2.$$

The conditional variance of the error term is therefore constant across different values of the explanatory variable.
Since

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{S_{xx}},$$

we obtain

$$\boxed{\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2}{S_{xx}}}.$$

For the intercept,

$$\boxed{\mathrm{Var}(\hat{\beta}_0 \mid X) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}.$$

These formulas show that greater variation in the explanatory variable reduces the variance of the estimated slope. In particular, a larger S_{xx} provides more information about the relationship between x and y.
The standard deviations of the estimators are

$$\mathrm{sd}(\hat{\beta}_1 \mid X) = \frac{\sigma}{\sqrt{S_{xx}}}$$

and

$$\mathrm{sd}(\hat{\beta}_0 \mid X) = \sigma \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}.$$

Since σ^2 is generally unknown, these quantities cannot be directly computed from the observed sample.

1.12 Estimating the Error Variance

The OLS residuals provide an estimator of the unobserved error variance. For simple linear regression, define

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} \hat{u}_i^2}{n-2} = \frac{\mathrm{SSR}}{n-2}.$$

The denominator is $n-2$, rather than n, because two parameters, β_0 and β_1, have been estimated from the sample.

Under the classical simple regression assumptions,

$$\boxed{\mathbb{E}(\hat{\sigma}^2 \mid X) = \sigma^2}.$$

The estimated standard deviation of the regression error is

$$\hat{\sigma} = \sqrt{\frac{\mathrm{SSR}}{n-2}}.$$

Replacing the unknown σ by $\hat{\sigma}$ gives the standard errors

$$\boxed{\mathrm{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}}$$

and

$$\boxed{\mathrm{se}(\hat{\beta}_0) = \hat{\sigma}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}}.$$

The distinction between the variance of an estimator and its standard error is important:

$$\mathrm{Var}(\hat{\beta}_j \mid X)$$

is a theoretical quantity involving the unknown population variance σ^2, while

$$\mathrm{se}(\hat{\beta}_j)$$

is an observable estimate of the standard deviation of the estimator.

The next chapter extends these ideas from one explanatory variable to the multiple linear regression model, where partial effects, multicollinearity, variance inflation, and the Gauss–Markov theorem can be studied in a unified framework.

Chapter 2

Multiple Linear Regression and Finite-Sample Properties

2.1 The Multiple Linear Regression Model

The multiple linear regression model extends the simple regression framework by allowing several explanatory variables to affect the dependent variable simultaneously.

Model 2.1.1 (Multiple Linear Regression Model)**.** The multiple linear regression model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u.$$

Here, y is the dependent variable, $x_1, \ldots, x_k$ are explanatory variables, $\beta_0, \ldots, \beta_k$ are unknown parameters, and u is the error term.

The interpretation of β_j is a *partial effect*. Holding all other explanatory variables fixed,

$$\Delta y = \beta_j \Delta x_j.$$

Thus β_j measures the effect of changing x_j while keeping $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$ constant.

Example 2.1.2 (Wage Equation)**.** Consider

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{tenure} + u.$$

Then β_1 measures the change in wage associated with one additional year of education, holding work experience and tenure fixed.
This interpretation is stronger than the coefficient from a simple regression of wage on education because some observable factors that also affect wage are now controlled for explicitly.

As in simple regression, the model is linear in the parameters. The regressors themselves may be nonlinear transformations. For example,

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

is still a multiple linear regression model.

2.2 OLS Estimation

Suppose the sample contains observations

$$(y_i, x_{i1}, \ldots, x_{ik}), \qquad i = 1, \ldots, n.$$

For arbitrary coefficients $b_0, \ldots, b_k$, define

$$e_i = y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik}.$$

The OLS estimator minimizes the sum of squared residuals:

$$(\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k) = \underset{b_0,\ldots,b_k}{\arg\min} \sum_{i=1}^{n} e_i^2.$$

Equivalently,

$$(\hat{\beta}_0, \ldots, \hat{\beta}_k) = \underset{b_0,\ldots,b_k}{\arg\min} \sum_{i=1}^{n} \left(y_i - b_0 - \sum_{j=1}^{k} b_j x_{ij} \right)^2.$$

The first-order conditions are

$$\sum_{i=1}^{n} \hat{u}_i = 0$$

and

$$\sum_{i=1}^{n} x_{ij}\hat{u}_i = 0, \qquad j = 1, \ldots, k,$$

where

$$\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}.$$

Therefore the residuals are orthogonal to every included explanatory variable.

2.3 Matrix Form of OLS

The multiple regression model can be written compactly as

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{u},$$

where

$$\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix},$$

and

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}.$$

OLS minimizes

$$\|\boldsymbol{y} - X\boldsymbol{b}\|^2.$$

The first-order condition is

$$X^\top(\boldsymbol{y} - X\hat{\boldsymbol{\beta}}) = 0,$$

so

$$X^\top X\hat{\boldsymbol{\beta}} = X^\top \boldsymbol{y}.$$

If $X^\top X$ is invertible, then

$$\boxed{\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \boldsymbol{y}}.$$

The fitted values are

$$\hat{\boldsymbol{y}} = X\hat{\boldsymbol{\beta}} = X(X^\top X)^{-1} X^\top \boldsymbol{y}.$$

Define the projection matrix

$$P_X = X(X^\top X)^{-1} X^\top.$$

Then

$$\hat{\boldsymbol{y}} = P_X \boldsymbol{y}.$$

The residual vector is

$$\hat{\boldsymbol{u}} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (I - P_X)\boldsymbol{y}.$$

Define

$$M_X = I - P_X.$$

Then

$$\boxed{\hat{\boldsymbol{u}} = M_X \boldsymbol{y}}.$$

Both P_X and M_X are symmetric and idempotent:

$$P_X^\top = P_X, \qquad P_X^2 = P_X,$$

and

$$M_X^\top = M_X, \qquad M_X^2 = M_X.$$

2.4 Partialling Out and the Frisch–Waugh–Lovell Result

One of the most useful interpretations of multiple regression is that the coefficient on x_j measures the relationship between y and the part of x_j that cannot be explained by the other regressors.
Consider

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u.$$

To isolate the effect of x_1, first regress x_1 on x_2:

$$x_1 = \gamma_0 + \gamma_1 x_2 + r_1.$$

The residual r_1 is the part of x_1 that is orthogonal to x_2. Then the multiple regression coefficient $\hat{\beta}_1$ is equal to the slope coefficient from regressing y on r_1:

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n r_{i1} y_i}{\sum_{i=1}^n r_{i1}^2}}.$$

This result is known as the *partialling out* principle or the Frisch–Waugh–Lovell theorem. More generally, let r_{ij} be the residual from regressing x_j on all other explanatory variables. Then

$$\boxed{\hat{\beta}_j = \frac{\sum_{i=1}^n r_{ij} y_i}{\sum_{i=1}^n r_{ij}^2}}.$$

Thus multiple regression removes the linear effects of the other regressors before estimating the effect of x_j.

Example 2.4.1 (Education Controlling for Experience)**.** Suppose

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + u.$$

First regress education on experience:

$$\text{educ} = \gamma_0 + \gamma_1 \text{exper} + r.$$

The residual r represents the component of education that is not linearly explained by experience. The coefficient $\hat{\beta}_1$ is then determined by the relationship between wage and this residual component of education.
This formalizes the phrase "holding experience fixed."

2.5 The Classical Linear Model Assumptions

The finite-sample properties of OLS depend on a collection of assumptions.

Assumption 2.5.1 (Linear in Parameters)**.** The population model is

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u.$$

Assumption 2.5.2 (Random Sampling)**.** The observations

$$(y_i, x_{i1}, \ldots, x_{ik}), \qquad i = 1, \ldots, n,$$

are randomly sampled from the population.

Assumption 2.5.3 (No Perfect Collinearity)**.** No explanatory variable is an exact linear combination of the others. Equivalently,

$$\operatorname{rank}(X) = k + 1.$$

This assumption guarantees that $X^\top X$ is invertible.
Perfect collinearity occurs, for example, if

$$x_3 = 2x_1 - 5x_2$$

for every observation.

Assumption 2.5.4 (Zero Conditional Mean)**.**

$$\boxed{\mathbb{E}(u \mid x_1, \ldots, x_k) = 0}.$$

This assumption implies that each included explanatory variable is uncorrelated with the error term:

$$\operatorname{Cov}(x_j, u) = 0, \qquad j = 1, \ldots, k.$$

A variable correlated with u is called *endogenous*, while a variable satisfying the relevant exogeneity condition is called *exogenous*.

Assumption 2.5.5 (Homoskedasticity)**.**

$$\boxed{\operatorname{Var}(u \mid x_1, \ldots, x_k) = \sigma^2}.$$

Under homoskedasticity,

$$\operatorname{Var}(y \mid x_1, \ldots, x_k) = \sigma^2.$$

2.6 Unbiasedness of OLS

Using

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{u},$$

the OLS estimator becomes

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top (X\boldsymbol{\beta} + \boldsymbol{u}).$$

Therefore

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (X^\top X)^{-1} X^\top \boldsymbol{u}.$$

Under the zero conditional mean assumption,

$$\mathbb{E}(\boldsymbol{u} \mid X) = 0.$$

Hence

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta} + (X^\top X)^{-1} X^\top \mathbb{E}(\boldsymbol{u} \mid X) = \boldsymbol{\beta}.$$

Therefore,

$$\boxed{\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta}}.$$

In particular,

$$\boxed{\mathbb{E}(\hat{\beta}_j) = \beta_j, \qquad j = 0, \ldots, k}.$$

Thus OLS is unbiased under the classical exogeneity assumptions.

2.7 Omitted Variable Bias

The zero conditional mean assumption may fail when an important variable is omitted from the regression.
Consider the population model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u.$$

Suppose x_2 is omitted and the estimated regression is

$$y = \alpha_0 + \alpha_1 x_1 + v.$$

Write the linear projection of x_2 on x_1 as

$$x_2 = \delta_0 + \delta_1 x_1 + r.$$

Substituting into the true model gives

$$y = (\beta_0 + \beta_2 \delta_0) + (\beta_1 + \beta_2 \delta_1) x_1 + (\beta_2 r + u).$$

Therefore the coefficient from the misspecified regression targets

$$\alpha_1 = \beta_1 + \beta_2 \delta_1.$$

The omitted variable bias is

$$\boxed{\alpha_1 - \beta_1 = \beta_2 \delta_1}.$$

Thus omitted variable bias occurs when both of the following conditions hold:

- the omitted variable affects y, so $\beta_2 \neq 0$;
- the omitted variable is correlated with the included regressor, so $\delta_1 \neq 0$.

Example 2.7.1 (Ability Bias in the Return to Education)**.** Suppose the true model is

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{ability} + u.$$

If ability increases wages, then $\beta_2 > 0$. If individuals with higher ability also obtain more education, education and ability are positively correlated, so $\delta_1 > 0$.
If ability is omitted, then

$$\alpha_1 - \beta_1 = \beta_2 \delta_1 > 0.$$

Hence the simple regression coefficient on education tends to overestimate the direct effect of education in this example.

2.8 Variance of the OLS Estimator

Under homoskedasticity,

$$\mathrm{Var}(\boldsymbol{u} \mid X) = \sigma^2 I_n.$$

Since

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{u},$$

we obtain

$$\begin{aligned}\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) &= (X^\top X)^{-1} X^\top \mathrm{Var}(\boldsymbol{u} \mid X) X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1}.\end{aligned}$$

Therefore,

$$\boxed{\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}}.$$

For a particular slope coefficient $\hat{\beta}_j$, an equivalent scalar formula is

$$\boxed{\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\mathrm{SST}_j (1 - R_j^2)}},$$

where

$$\mathrm{SST}_j = \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2,$$

and R_j^2 is the R^2 from regressing x_j on all other explanatory variables.
This formula identifies three determinants of the variance of $\hat{\beta}_j$:

- the error variance σ^2;
- the sample variation in x_j;
- the degree of linear dependence between x_j and the remaining regressors.

A larger error variance increases estimator variance, while greater independent variation in x_j decreases estimator variance.

2.9 Multicollinearity and Variance Inflation

When x_j is strongly linearly related to the other explanatory variables, the auxiliary regression R_j^2 is close to 1. Then

$$1 - R_j^2$$

is small, and

$$\mathrm{Var}(\hat{\beta}_j \mid X)$$

becomes large.
This phenomenon is called *multicollinearity*.
Perfect multicollinearity makes OLS impossible because $X^\top X$ is singular. Imperfect multicollinearity does not make OLS biased, but it can substantially increase standard errors.

Definition 2.9.1 (Variance Inflation Factor)**.** The variance inflation factor for x_j is

$$\boxed{\mathrm{VIF}_j = \frac{1}{1 - R_j^2}}.$$

Thus

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\mathrm{SST}_j}\,\mathrm{VIF}_j\,.$$

A large VIF_j indicates that the variance of $\hat{\beta}_j$ is inflated because x_j contains relatively little variation that is independent of the other regressors.

Example 2.9.2 (Age and Experience)**.** Suppose a wage regression includes both age and work experience:

$$\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{age} + \beta_3 \text{exper} + u.$$

Age and experience are typically strongly correlated. This does not necessarily bias the OLS estimators, but it may make it difficult to separately estimate the effects of age and experience, resulting in large standard errors for $\hat{\beta}_2$ and $\hat{\beta}_3$.

2.10 Estimating the Error Variance

Let

$$\hat{\boldsymbol{u}} = \boldsymbol{y} - X\hat{\boldsymbol{\beta}}.$$

The residual sum of squares is

$$\mathrm{SSR} = \hat{\boldsymbol{u}}^\top \hat{\boldsymbol{u}} = \sum_{i=1}^{n} \hat{u}_i^2.$$

Since $k+1$ regression coefficients are estimated, the residual degrees of freedom are

$$n - k - 1.$$

The estimator of the error variance is

$$\boxed{\hat{\sigma}^2 = \frac{\mathrm{SSR}}{n-k-1}}.$$

To see why the degrees-of-freedom correction appears, recall that

$$\hat{\boldsymbol{u}} = M_X \boldsymbol{y}.$$

Since

$$M_X X = 0,$$

we have

$$\hat{\boldsymbol{u}} = M_X \boldsymbol{u}.$$

Therefore,

$$\hat{\boldsymbol{u}}^\top \hat{\boldsymbol{u}} = \boldsymbol{u}^\top M_X \boldsymbol{u}.$$

Using the trace identity

$$a^\top B a = \mathrm{tr}(B a a^\top),$$

we obtain

$$\begin{aligned}\mathbb{E}(\hat{\boldsymbol{u}}^\top \hat{\boldsymbol{u}} \mid X) &= \mathbb{E}(\boldsymbol{u}^\top M_X \boldsymbol{u} \mid X) \\ &= \mathrm{tr}\left(M_X \mathbb{E}(\boldsymbol{u}\boldsymbol{u}^\top \mid X)\right).\end{aligned}$$

Under homoskedasticity,

$$\mathbb{E}(\boldsymbol{u}\boldsymbol{u}^\top \mid X) = \sigma^2 I_n.$$

Hence

$$\begin{aligned}\mathbb{E}(\hat{\boldsymbol{u}}^\top \hat{\boldsymbol{u}} \mid X) &= \sigma^2 \operatorname{tr}(M_X) \\ &= \sigma^2(n-k-1).\end{aligned}$$

Therefore,

$$\boxed{\mathbb{E}(\hat{\sigma}^2 \mid X) = \sigma^2}.$$

2.11 Standard Errors in Multiple Regression

Because

$$\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1},$$

the unknown σ^2 is replaced by $\hat{\sigma}^2$.
Thus the estimated covariance matrix is

$$\boxed{\widehat{\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X)} = \hat{\sigma}^2 (X^\top X)^{-1}}.$$

If the j-th diagonal element of $(X^\top X)^{-1}$ is denoted by c_{jj}, then

$$\operatorname{se}(\hat{\beta}_j) = \hat{\sigma}\sqrt{c_{jj}}.$$

Equivalently,

$$\boxed{\operatorname{se}(\hat{\beta}_j) = \frac{\hat{\sigma}}{\sqrt{\operatorname{SST}_j(1-R_j^2)}}}.$$

This formula shows directly how multicollinearity increases standard errors.

2.12 The Gauss–Markov Theorem

Unbiasedness alone does not determine a unique estimator. Many estimators may satisfy

$$\mathbb{E}(\tilde{\beta}_j) = \beta_j.$$

The Gauss–Markov theorem establishes an important optimality property of OLS.

Definition 2.12.1 (BLUE)**.** An estimator is called the *Best Linear Unbiased Estimator* if it is

- linear in the observed dependent variables;
- unbiased;
- of minimum variance among all linear unbiased estimators.

Theorem 2.12.2 (Gauss–Markov Theorem)**.** Under the classical linear model assumptions, in particular zero conditional mean and homoskedasticity, the OLS estimators are BLUE.

To illustrate the idea, consider an arbitrary linear unbiased estimator of β_j:

$$\tilde{\beta}_j = \sum_{i=1}^{n} w_i y_i,$$

where the weights w_i may depend on the explanatory variables.

Unbiasedness imposes restrictions on the weights. The OLS estimator uses a particular set of weights satisfying these restrictions and, under homoskedasticity, minimizes the conditional variance.
Hence, for any other linear unbiased estimator $\tilde{\beta}_j$,

$$\boxed{\mathrm{Var}(\tilde{\beta}_j \mid X) \geq \mathrm{Var}(\hat{\beta}_j \mid X)}.$$

The word "best" refers specifically to minimum variance within the class of linear unbiased estimators. It does not imply that OLS is optimal among every possible estimator under every possible model.

2.13 What the Classical Assumptions Guarantee

The role of the assumptions can now be summarized systematically.
Under linearity in parameters, random sampling, no perfect collinearity, and zero conditional mean,

$$\boxed{\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta}}.$$

Adding homoskedasticity gives

$$\boxed{\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}}$$

and the Gauss–Markov result that OLS is BLUE.
These results concern the finite-sample behavior of OLS. They do not yet provide exact probability distributions for the estimators.
To construct finite-sample confidence intervals and hypothesis tests, an additional distributional assumption on the regression errors is introduced in the next chapter.

Chapter 3

Statistical Inference in Linear Regression

3.1 Normality of the Error Term

The finite-sample properties derived in the previous chapter establish unbiasedness and variance formulas for OLS. To obtain exact finite-sample distributions and construct hypothesis tests, an additional distributional assumption is introduced.

Assumption 3.1.1 (Normality)**.** Conditional on the explanatory variables,

$$u \mid X \sim N(0, \sigma^2 I_n).$$

Equivalently, for each observation,

$$u_i \mid x_{i1}, \ldots, x_{ik} \sim N(0, \sigma^2).$$

Under this assumption, the multiple regression model

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{u}$$

implies

$$\boxed{\boldsymbol{y} \mid X \sim N(X\boldsymbol{\beta}, \sigma^2 I_n)}.$$

Normality is not required for OLS to be unbiased or BLUE. Its role here is to provide exact finite-sample distributions for the estimators and test statistics.

3.2 Normal Distribution Results Used in Regression

Only a few standard properties of multivariate normal distributions are needed for regression inference.
If

$$\boldsymbol{Y} \sim N(\boldsymbol{\mu}, \Sigma),$$

then every component Y_i is normally distributed.
For any constant matrix A and vector $\boldsymbol{b}$,

$$A\boldsymbol{Y} + \boldsymbol{b} \sim N(A\boldsymbol{\mu} + \boldsymbol{b}, A\Sigma A^\top).$$

For jointly normally distributed random vectors, zero covariance implies independence. Thus, if

$$A\Sigma B^\top = 0,$$

then $A\boldsymbol{Y}$ and $B\boldsymbol{Y}$ are independent.
Finally, if

$$\boldsymbol{z} \sim N(0, I_n)$$

and A is symmetric and idempotent with

$$\operatorname{rank}(A) = r,$$

then

$$\boxed{\boldsymbol{z}^\top A \boldsymbol{z} \sim \chi^2_r}.$$

These facts will be used repeatedly in deriving the distributions of OLS estimators and residual sums of squares.

3.3 Distribution of the OLS Estimator

Recall that

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \boldsymbol{y}.$$

Conditional on X,

$$\boldsymbol{y} \sim N(X\boldsymbol{\beta}, \sigma^2 I_n).$$

Since $\hat{\boldsymbol{\beta}}$ is a linear transformation of $\boldsymbol{y}$,

$$\begin{aligned}\hat{\boldsymbol{\beta}} \mid X &\sim N\left((X^\top X)^{-1} X^\top X\boldsymbol{\beta}, \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1}\right) \\ &= N\left(\boldsymbol{\beta}, \sigma^2 (X^\top X)^{-1}\right).\end{aligned}$$

Therefore,

$$\boxed{\hat{\boldsymbol{\beta}} \mid X \sim N\left(\boldsymbol{\beta}, \sigma^2 (X^\top X)^{-1}\right)}.$$

For an individual coefficient,

$$\boxed{\hat{\beta}_j \mid X \sim N\left(\beta_j, \mathrm{Var}(\hat{\beta}_j \mid X)\right)}.$$

If σ were known, then

$$\frac{\hat{\beta}_j - \beta_j}{\mathrm{sd}(\hat{\beta}_j \mid X)} \sim N(0,1).$$

In practice, however, σ^2 is unknown and must be estimated.

3.4 Distribution of the Error Variance Estimator

Recall that

$$\hat{\sigma}^2 = \frac{\mathrm{SSR}}{n-k-1},$$

where

$$\mathrm{SSR} = \hat{\boldsymbol{u}}^\top \hat{\boldsymbol{u}}.$$

Since

$$\hat{\boldsymbol{u}} = M_X \boldsymbol{u},$$

we have

$$\mathrm{SSR} = \boldsymbol{u}^\top M_X \boldsymbol{u}.$$

Define

$$\boldsymbol{z} = \frac{\boldsymbol{u}}{\sigma}.$$

Under the normality assumption,

$$\boldsymbol{z} \mid X \sim N(0, I_n).$$

Because M_X is symmetric and idempotent and

$$\mathrm{rank}(M_X) = n - k - 1,$$

it follows that

$$\frac{\mathrm{SSR}}{\sigma^2} = \boldsymbol{z}^\top M_X \boldsymbol{z} \sim \chi^2_{n-k-1}.$$

Therefore,

$$\boxed{\frac{(n-k-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-k-1}}.$$

Moreover, under normality,

$$\hat{\boldsymbol{\beta}} \quad \text{and} \quad \hat{\sigma}^2$$

are independent.
This independence is the key fact that leads to the Student t-distribution.

3.5 The t-Distribution for OLS Coefficients

Let

$$\text{se}(\hat{\beta}_j)$$

denote the estimated standard error of $\hat{\beta}_j$.
Since

$$\frac{\hat{\beta}_j - \beta_j}{\text{sd}(\hat{\beta}_j \mid X)} \sim N(0,1)$$

and

$$\frac{(n-k-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-k-1}$$

independently, we obtain

$$\boxed{\frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \sim t_{n-k-1}}.$$

This result is the foundation of exact finite-sample inference for individual regression coefficients.

3.6 Testing a Single Regression Coefficient

Suppose the null hypothesis is

$$H_0 : \beta_j = c,$$

where c is a specified constant.
The corresponding t-statistic is

$$\boxed{t = \frac{\hat{\beta}_j - c}{\text{se}(\hat{\beta}_j)}}.$$

Under H_0,

$$t \sim t_{n-k-1}.$$

The most common case is

$$H_0 : \beta_j = 0,$$

which tests whether x_j has a statistically detectable partial relationship with y.

3.7 One-Sided and Two-Sided Tests

For the right-tailed alternative

$$H_0 : \beta_j = 0, \qquad H_1 : \beta_j > 0,$$

reject H_0 at significance level α when

$$t > t_{\alpha,n-k-1},$$

where $t_{\alpha,n-k-1}$ denotes the appropriate upper-tail critical value.
For the left-tailed alternative

$$H_0 : \beta_j = 0, \qquad H_1 : \beta_j < 0,$$

reject H_0 when

$$t < -t_{\alpha,n-k-1}.$$

For the two-sided alternative

$$H_0 : \beta_j = 0, \qquad H_1 : \beta_j \neq 0,$$

reject H_0 when

$$|t| > t_{\alpha/2,n-k-1}.$$

Typical significance levels are $\alpha = 0.10$, 0.05, and 0.01.

Example 3.7.1 (Testing the Return to Education)**.** Consider

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + u.$$

Suppose

$$\hat{\beta}_1 = 0.075, \qquad \text{se}(\hat{\beta}_1) = 0.018.$$

To test

$$H_0 : \beta_1 = 0 \qquad \text{against} \qquad H_1 : \beta_1 > 0,$$

the test statistic is

$$t = \frac{0.075}{0.018} \approx 4.17.$$

A sufficiently large positive t-statistic provides evidence against the null hypothesis and in favor of a positive partial association between education and wage.

3.8 p-Values

Instead of comparing a test statistic with a critical value, inference can be expressed using a p-value.
For a two-sided test,

$$H_0 : \beta_j = c, \qquad H_1 : \beta_j \neq c,$$

the p-value is

$$p = 2\mathbb{P}\left(T_{n-k-1} > |t_{\text{obs}}|\right).$$

The null hypothesis is rejected at significance level α if $p < \alpha$.
A small p-value indicates that the observed test statistic would be unlikely if the null hypothesis were true. It does not measure the probability that the null hypothesis itself is true.

3.9 Confidence Intervals

A two-sided $100(1-\alpha)\%$ confidence interval for β_j is

$$\boxed{\hat{\beta}_j \pm t_{\alpha/2,n-k-1}\,\mathrm{se}(\hat{\beta}_j)}.$$

Equivalently,

$$\mathbb{P}\left(\hat{\beta}_j - t_{\alpha/2,n-k-1}\,\mathrm{se}(\hat{\beta}_j) \le \beta_j \le \hat{\beta}_j + t_{\alpha/2,n-k-1}\,\mathrm{se}(\hat{\beta}_j)\right) = 1-\alpha.$$

For the two-sided test

$$H_0 : \beta_j = c,$$

the null is rejected at significance level α exactly when c lies outside the corresponding $100(1-\alpha)\%$ confidence interval.

3.10 Testing Linear Combinations of Parameters

Regression hypotheses do not always concern a single coefficient.
For example,

$$H_0 : \beta_1 - \beta_2 = 0$$

is equivalent to

$$H_0 : \beta_1 = \beta_2.$$

Define

$$\theta = \beta_1 - \beta_2.$$

Then

$$\hat{\theta} = \hat{\beta}_1 - \hat{\beta}_2.$$

Its variance is

$$\begin{aligned}\mathrm{Var}(\hat{\theta} \mid X) = \mathrm{Var}(\hat{\beta}_1 \mid X) + \mathrm{Var}(\hat{\beta}_2 \mid X) \\ - 2\,\mathrm{Cov}(\hat{\beta}_1, \hat{\beta}_2 \mid X).\end{aligned}$$

The corresponding test statistic is

$$t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{\mathrm{se}(\hat{\beta}_1 - \hat{\beta}_2)}.$$

More generally, a single linear restriction can be written as

$$H_0 : \boldsymbol{a}^\top \boldsymbol{\beta} = c.$$

Then

$$t = \frac{\boldsymbol{a}^\top \hat{\boldsymbol{\beta}} - c}{\mathrm{se}(\boldsymbol{a}^\top \hat{\boldsymbol{\beta}})}.$$

3.11 Joint Hypothesis Testing

A t-test handles one linear restriction. When several restrictions must be tested simultaneously, an F-test is used.
Suppose the unrestricted model is

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u.$$

Consider q restrictions summarized as

$$H_0 : R\boldsymbol{\beta} = \boldsymbol{r}.$$

The unrestricted model is estimated without imposing these restrictions, giving residual sum of squares SSR_{ur}.
The restricted model imposes H_0, giving SSR_r.
Since the unrestricted model can fit the data at least as well,

$$\text{SSR}_r \geq \text{SSR}_{ur}.$$

The F-statistic is

$$\boxed{F = \frac{(\text{SSR}_r - \text{SSR}_{ur})/q}{\text{SSR}_{ur}/(n-k-1)}}.$$

Under the null hypothesis,

$$\boxed{F \sim F_{q,n-k-1}}.$$

Large values of F indicate that imposing the restrictions causes a substantial deterioration in fit and therefore provide evidence against H_0.

3.12 Testing Joint Significance

A common application is to test whether several explanatory variables are jointly irrelevant.
Suppose

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u.$$

To test whether x_2 and x_3 have no explanatory power conditional on x_1, consider

$$H_0 : \beta_2 = \beta_3 = 0.$$

The restricted model becomes

$$y = \beta_0 + \beta_1 x_1 + u.$$

Here there are $q = 2$ restrictions, so

$$F = \frac{(\text{SSR}_r - \text{SSR}_{ur})/2}{\text{SSR}_{ur}/(n-k-1)}.$$

If H_0 is rejected, x_2 and x_3 are said to be *jointly statistically significant.*
Individual t-tests and a joint F-test answer different questions. It is possible for individual coefficients to be insignificant while the variables are jointly significant.

3.13 The R^2 Form of the F-Test

When the restricted and unrestricted models have the same dependent variable and both contain an intercept, the F-statistic can also be written using R^2.
Let R^2_{ur} and R^2_r denote the coefficients of determination from the unrestricted and restricted regressions. Then

$$\boxed{F = \frac{(R^2_{ur} - R^2_r)/q}{(1 - R^2_{ur})/(n-k-1)}}.$$

This form is often convenient when regression output reports R^2 but not the residual sums of squares.

3.14 The Overall Significance of a Regression

A special joint test concerns all slope coefficients:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0.$$

The restricted model contains only an intercept:

$$y = \beta_0 + u.$$

There are k restrictions. Since the restricted model has $R_r^2 = 0$, the F-statistic reduces to

$$\boxed{F = \frac{R^2/k}{(1 - R^2)/(n - k - 1)}}.$$

Rejecting the null means that the explanatory variables are jointly statistically significant.

3.15 Relationship Between t-Tests and F-Tests

When the null hypothesis contains only one restriction, the F-test and the corresponding two-sided t-test are equivalent.
In particular,

$$\boxed{F = t^2}$$

when $q = 1$.
Thus the F-test can be viewed as a generalization of the t-test to multiple simultaneous linear restrictions.

3.16 Statistical and Practical Significance

Statistical significance and practical significance are different concepts.
A coefficient may be statistically significant because its standard error is small, even when the estimated effect itself is economically or practically negligible.
Conversely, an economically large coefficient may fail to be statistically significant when the estimator is imprecise.
Thus inference should consider both

$$\text{estimated magnitude}$$

and

$$\text{statistical uncertainty.}$$

Example 3.16.1 (A Small but Precisely Estimated Effect)**.** Suppose a regression estimates

$$\hat{\beta}_1 = 0.001$$

with

$$\text{se}(\hat{\beta}_1) = 0.0002.$$

Then

$$t = \frac{0.001}{0.0002} = 5.$$

The coefficient is highly statistically significant, but whether an effect of 0.001 is practically important depends on the units and the application. Statistical significance alone cannot answer that question.

3.17 Inference for the Conditional Mean

Suppose the objective is to estimate the conditional mean of y at particular regressor values

$$\boldsymbol{x}_0 = \begin{pmatrix} 1 \\ c_1 \\ \vdots \\ c_k \end{pmatrix}.$$

The conditional mean is

$$\mathbb{E}(y \mid \boldsymbol{x}_0) = \boldsymbol{x}_0^\top \boldsymbol{\beta},$$

and its estimator is

$$\hat{y}_0 = \boldsymbol{x}_0^\top \hat{\boldsymbol{\beta}}.$$

Its estimated variance is

$$\widehat{\mathrm{Var}(\hat{y}_0 \mid X)} = \hat{\sigma}^2 \boldsymbol{x}_0^\top (X^\top X)^{-1} \boldsymbol{x}_0.$$

Hence

$$\mathrm{se}(\hat{y}_0) = \hat{\sigma} \sqrt{\boldsymbol{x}_0^\top (X^\top X)^{-1} \boldsymbol{x}_0}.$$

A $100(1-\alpha)\%$ confidence interval for the conditional mean is

$$\boxed{\hat{y}_0 \pm t_{\alpha/2, n-k-1} \, \mathrm{se}(\hat{y}_0)}.$$

3.18 Prediction Intervals

Estimating the conditional mean and predicting a new observation are different problems. For a new observation,

$$y_0 = \boldsymbol{x}_0^\top \boldsymbol{\beta} + u_0.$$

The prediction error is

$$y_0 - \hat{y}_0 = u_0 + \boldsymbol{x}_0^\top (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}).$$

Therefore,

$$\mathrm{Var}(y_0 - \hat{y}_0 \mid X) = \sigma^2 \left[1 + \boldsymbol{x}_0^\top (X^\top X)^{-1} \boldsymbol{x}_0 \right].$$

The estimated standard error of prediction is

$$\boxed{\mathrm{se}_{\mathrm{pred}} = \hat{\sigma} \sqrt{1 + \boldsymbol{x}_0^\top (X^\top X)^{-1} \boldsymbol{x}_0}}.$$

Thus a $100(1-\alpha)\%$ prediction interval is

$$\boxed{\hat{y}_0 \pm t_{\alpha/2, n-k-1} \, \mathrm{se}_{\mathrm{pred}}}.$$

The prediction interval is wider than the confidence interval for the conditional mean because predicting an individual outcome involves the additional uncertainty from the new error term u_0.

3.19 Summary of Finite-Sample Inference

Under the classical linear model assumptions and conditional normality,

$$\hat{\boldsymbol{\beta}} \mid X \sim N\left(\boldsymbol{\beta}, \sigma^2 (X^\top X)^{-1}\right),$$

while

$$\frac{(n-k-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-k-1}.$$

These two quantities are independent. Consequently,

$$\frac{\hat{\beta}_j - \beta_j}{\mathrm{se}(\hat{\beta}_j)} \sim t_{n-k-1},$$

which provides exact inference for individual coefficients.
For several simultaneous linear restrictions,

$$H_0 : R\boldsymbol{\beta} = \boldsymbol{r},$$

the corresponding statistic follows an F-distribution under the null hypothesis.
The finite-sample theory developed in this chapter relies strongly on the normality assumption. The next chapter replaces exact finite-sample distributions with large-sample approximations and develops consistency and asymptotic normality of OLS.

Chapter 4

Asymptotic Theory of OLS

4.1 Why Asymptotic Theory Is Needed

The exact finite-sample inference developed in the previous chapter relies on the normality assumption for the regression errors. In many applications, however, exact normality is difficult to justify.

Asymptotic theory studies the behavior of estimators as the sample size n becomes large. Instead of requiring an exact finite-sample distribution, it provides approximate results based on probability limits, laws of large numbers, and central limit theorems.

The two central questions are:

- Does the estimator converge to the true parameter?
- What is the limiting distribution of the estimation error?

These correspond respectively to *consistency* and *asymptotic normality*.

4.2 Modes of Convergence

Several notions of convergence appear in asymptotic analysis.

Definition 4.2.1 (Convergence in Probability)**.** A sequence X_n converges in probability to X, written

$$X_n \xrightarrow{p} X,$$

if for every $\varepsilon > 0$,

$$\mathbb{P}\left(|X_n - X| > \varepsilon\right) \to 0$$

as $n \to \infty$.

Equivalently,

$$\mathbb{P}\left(|X_n - X| < \varepsilon\right) \to 1.$$

Definition 4.2.2 (Almost Sure Convergence)**.** A sequence X_n converges almost surely to X, written

$$X_n \xrightarrow{a.s.} X,$$

if

$$\mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1.$$

Almost sure convergence is stronger than convergence in probability:

$$X_n \xrightarrow{a.s.} X \quad \Longrightarrow \quad X_n \xrightarrow{p} X.$$

Definition 4.2.3 (Convergence in Distribution)**.** A sequence X_n converges in distribution to X, written

$$X_n \xrightarrow{d} X,$$

if

$$F_{X_n}(x) \to F_X(x)$$

at every continuity point of F_X.

In regression asymptotics, convergence in probability is primarily used to establish consistency, while convergence in distribution is used to derive limiting distributions.

4.3 Probability Limits and Continuous Transformations

The probability limit of a sequence X_n is written

$$\text{plim}_{n\to\infty} X_n = X$$

when

$$X_n \xrightarrow{p} X.$$

Probability limits satisfy rules analogous to ordinary limits. If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then

$$X_n + Y_n \xrightarrow{p} X + Y,$$
$$X_n Y_n \xrightarrow{p} XY,$$

and, provided $Y \neq 0$,

$$\frac{X_n}{Y_n} \xrightarrow{p} \frac{X}{Y}.$$

More generally, if g is continuous and $X_n \xrightarrow{p} X$, then

$$\boxed{g(X_n) \xrightarrow{p} g(X)}.$$

This is the continuous mapping principle.

4.4 Law of Large Numbers

Suppose $X_1, \ldots, X_n$ are i.i.d. with finite mean $\mathbb{E}(X_i) = \mu$. The sample mean is

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

The law of large numbers implies

$$\boxed{\bar{X}_n \xrightarrow{p} \mu}.$$

Under stronger conditions, the strong law gives

$$\bar{X}_n \xrightarrow{a.s.} \mu.$$

The law of large numbers is the main tool used to replace sample moments by their corresponding population moments in large samples.
For example,

$$\frac{1}{n}\sum_{i=1}^{n} x_i^2 \xrightarrow{p} \mathbb{E}(x^2)$$

and

$$\frac{1}{n}\sum_{i=1}^{n} x_i u_i \xrightarrow{p} \mathbb{E}(xu).$$

4.5 Central Limit Theorem

Suppose $X_1, \ldots, X_n$ are i.i.d. with $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then

$$\boxed{\sqrt{n}\left(\bar{X}_n - \mu\right) \xrightarrow{d} N(0, \sigma^2)}.$$

Equivalently,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1).$$

The law of large numbers explains why estimators converge to their true values, whereas the central limit theorem explains the scale and shape of their remaining estimation error.

4.6 Consistency

Definition 4.6.1 (Consistency)**.** An estimator $\hat{\theta}_n$ is consistent for θ if

$$\boxed{\hat{\theta}_n \xrightarrow{p} \theta}.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\hat{\theta}_n - \theta\right| < \varepsilon\right) \to 1.$$

Consistency is different from unbiasedness.
Unbiasedness is the finite-sample property

$$\mathbb{E}(\hat{\theta}_n) = \theta,$$

whereas consistency concerns the limiting behavior

$$\hat{\theta}_n \xrightarrow{p} \theta.$$

An estimator may be biased in finite samples but still consistent if the bias disappears as $n \to \infty$.

4.7 Consistency of the Simple OLS Estimator

Consider the simple regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

The OLS slope estimator can be written as

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Dividing numerator and denominator by n,

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n}\sum_{i=1}^n (x_i - \bar{x})u_i}{\frac{1}{n}\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Under suitable random sampling and moment conditions,

$$\frac{1}{n}\sum_{i=1}^n (x_i - \bar{x})u_i \xrightarrow{p} \mathrm{Cov}(x, u),$$

while

$$\frac{1}{n}\sum_{i=1}^n (x_i - \bar{x})^2 \xrightarrow{p} \mathrm{Var}(x).$$

Therefore,

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \frac{\mathrm{Cov}(x, u)}{\mathrm{Var}(x)}.$$

If

$$\mathrm{Cov}(x, u) = 0,$$

then

$$\boxed{\operatorname{plim} \hat{\beta}_1 = \beta_1}.$$

Hence the OLS slope estimator is consistent.

4.8 Asymptotic Bias under Endogeneity

The probability-limit representation

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \frac{\operatorname{Cov}(x, u)}{\operatorname{Var}(x)}$$

also explains why endogeneity is a fundamental problem.
If

$$\operatorname{Cov}(x, u) \neq 0,$$

then

$$\operatorname{plim} \hat{\beta}_1 \neq \beta_1.$$

Thus increasing the sample size does not eliminate the error caused by endogeneity.
The asymptotic bias is

$$\boxed{\operatorname{plim} \hat{\beta}_1 - \beta_1 = \frac{\operatorname{Cov}(x, u)}{\operatorname{Var}(x)}}.$$

Example 4.8.1 (Education and Ability)**.** Suppose

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + u,$$

where unobserved ability is contained in u.
If education and ability are positively correlated, then

$$\operatorname{Cov}(\text{educ}, u) > 0.$$

Consequently,

$$\operatorname{plim} \hat{\beta}_1 > \beta_1.$$

Collecting more observations does not solve this problem because the source of inconsistency is the relationship between the explanatory variable and the error term rather than sampling variability.

4.9 Consistency in Multiple Regression

For the multiple regression model

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{u},$$

the OLS estimator satisfies

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (X^\top X)^{-1} X^\top \boldsymbol{u}.$$

Multiplying numerator and denominator by $1/n$,

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \left(\frac{X^\top X}{n}\right)^{-1} \left(\frac{X^\top \boldsymbol{u}}{n}\right).$$

Suppose

$$\frac{X^\top X}{n} \xrightarrow{p} Q,$$

where Q is finite and nonsingular, and

$$\frac{X^\top \boldsymbol{u}}{n} \xrightarrow{p} 0.$$

Then

$$\boxed{\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}}.$$

Thus the essential large-sample exogeneity condition is that the sample covariance between the regressors and the error term converges to zero.

4.10 Asymptotic Normality of OLS

Consistency establishes that

$$\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta},$$

but does not describe the distribution of the estimation error.
To obtain a limiting distribution, consider

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}).$$

From the OLS formula,

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left(\frac{X^\top X}{n}\right)^{-1} \left(\frac{X^\top \boldsymbol{u}}{\sqrt{n}}\right).$$

Under suitable regularity conditions,

$$\frac{X^\top X}{n} \xrightarrow{p} Q$$

and, by a central limit theorem,

$$\frac{X^\top \boldsymbol{u}}{\sqrt{n}} \xrightarrow{d} N(0, \Omega).$$

Therefore,

$$\boxed{\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(0, Q^{-1}\Omega Q^{-1})}.$$

Under homoskedasticity,

$$\Omega = \sigma^2 Q,$$

so the expression simplifies to

$$\boxed{\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(0, \sigma^2 Q^{-1})}.$$

4.11 Asymptotic Normality of an Individual Coefficient

For an individual regression coefficient,

$$\boxed{\sqrt{n}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, \sigma^2/a_j^2)},$$

where the precise form of a_j^2 depends on the variation in x_j after controlling for the other explanatory variables.
Let r_{ij} denote the residual from regressing x_j on all other regressors. Then a useful large-sample quantity is

$$a_j^2 = \operatorname{plim}\left(\frac{1}{n}\sum_{i=1}^{n} r_{ij}^2\right).$$

Thus the asymptotic variance decreases when x_j contains more variation that cannot be explained by the remaining regressors.

4.12 Consistency of the Error Variance Estimator

Recall that

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} \hat{u}_i^2}{n-k-1}.$$

Under the usual large-sample assumptions and homoskedasticity,

$$\boxed{\hat{\sigma}^2 \xrightarrow{p} \sigma^2}.$$

Thus $\hat{\sigma}^2$ is a consistent estimator of

$$\sigma^2 = \mathrm{Var}(u).$$

Consequently, estimated standard errors can replace unknown population standard deviations in large-sample inference.

4.13 Asymptotic t-Statistics

Suppose

$$\sqrt{n}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, V_j)$$

and the estimated variance is consistent.
Then

$$\boxed{\frac{\hat{\beta}_j - \beta_j}{\mathrm{se}(\hat{\beta}_j)} \xrightarrow{d} N(0,1)}.$$

For testing

$$H_0 : \beta_j = c,$$

the large-sample test statistic is therefore

$$t = \frac{\hat{\beta}_j - c}{\mathrm{se}(\hat{\beta}_j)}.$$

For sufficiently large n, the statistic can be compared with standard normal critical values:

$$1.645, \qquad 1.96, \qquad 2.576$$

for common one- or two-sided significance levels.
The notation t-statistic is still commonly used even though its large-sample limiting distribution is standard normal rather than an exact Student t-distribution.

4.14 Finite-Sample and Asymptotic Inference

There are two distinct routes to regression inference.
Under conditional normality,

$$\frac{\hat{\beta}_j - \beta_j}{\mathrm{se}(\hat{\beta}_j)} \sim t_{n-k-1}$$

exactly in finite samples.
Without exact normality but under suitable regularity conditions,

$$\frac{\hat{\beta}_j - \beta_j}{\mathrm{se}(\hat{\beta}_j)} \xrightarrow{d} N(0,1).$$

Thus asymptotic theory allows approximately valid inference under weaker distributional assumptions when the sample size is sufficiently large.

4.15 The Lagrange Multiplier Test

The Lagrange multiplier test provides another method for testing restrictions in regression models.
The general idea is to estimate only the restricted model under the null hypothesis and then examine whether the residuals contain systematic information associated with the excluded variables.
In many regression applications, the LM statistic takes the form

$$\boxed{LM = nR^2},$$

where R^2 is obtained from an appropriate auxiliary regression.
Under the null hypothesis and suitable regularity conditions,

$$\boxed{LM \xrightarrow{d} \chi^2_q},$$

where q is the number of restrictions.
Therefore the null hypothesis is rejected for sufficiently large values of nR^2.
The exact construction of the auxiliary regression depends on the hypothesis being tested.
This form will appear again in tests for heteroskedasticity and serial correlation.

4.16 Wald, LM, and F-Type Tests

Several hypothesis-testing procedures are asymptotically equivalent.
A Wald-type test estimates the unrestricted model and measures how far the unrestricted estimate lies from the restrictions imposed by the null hypothesis.
An LM test estimates the restricted model and examines whether relaxing the restrictions would substantially improve the model.
Likelihood-ratio tests compare the maximized objective functions under the restricted and unrestricted models.
Under standard regularity conditions and large samples, these tests have the same asymptotic rejection behavior, even though their finite-sample values need not be identical.
For linear regression, the familiar F-test can often be interpreted as a finite-sample counterpart of these general large-sample testing principles.

4.17 Asymptotic Efficiency of OLS

Efficiency can also be studied in a large-sample setting.
Suppose two consistent estimators satisfy

$$\sqrt{n}(\hat{\theta}_1 - \theta) \xrightarrow{d} N(0, V_1)$$

and

$$\sqrt{n}(\hat{\theta}_2 - \theta) \xrightarrow{d} N(0, V_2).$$

The estimator with smaller asymptotic variance is said to be asymptotically more efficient.
Under the classical homoskedastic regression assumptions, OLS has the familiar efficiency property associated with the Gauss–Markov framework. In large samples, efficiency comparisons are expressed using the asymptotic covariance matrices of competing estimators.
A smaller asymptotic variance means that the estimator concentrates more rapidly around the true parameter as the sample size grows.

4.18 Summary of OLS Asymptotics

The large-sample theory of OLS can be summarized by two central results.
First, under suitable exogeneity and regularity conditions,

$$\boxed{\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}}.$$

Thus OLS is consistent.
Second,

$$\boxed{\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(0, V)},$$

for an appropriate asymptotic covariance matrix V.
Consequently,

$$\boxed{\frac{\hat{\beta}_j - \beta_j}{\operatorname{se}(\hat{\beta}_j)} \xrightarrow{d} N(0,1)}.$$

These results provide the basis for large-sample confidence intervals and hypothesis tests without requiring exact normality of the regression errors.
The remaining chapters consider situations in which the basic regression model must be modified, either because the functional specification changes or because assumptions such as homoskedasticity, independence, or exogeneity fail.

Chapter 5

Model Specification and Regression Design

5.1 Scale Changes and Standardized Coefficients

The numerical value of a regression coefficient depends on the units in which the variables are measured. Changing units changes the estimated coefficient, but does not change the underlying statistical information.
Suppose

$$y = \beta_0 + \beta_1 x + u.$$

If the dependent variable is multiplied by a constant a, so that $y^* = ay$, then

$$y^* = a\beta_0 + a\beta_1 x + au.$$

Thus the slope coefficient becomes $a\beta_1$. Its standard error is also multiplied by a, so the corresponding t-statistic remains unchanged.
Similarly, rescaling an explanatory variable changes the numerical value of its coefficient but does not change the statistical content of the regression.
To remove dependence on measurement units, standardized variables may be used.
Define

$$z_y = \frac{y - \bar{y}}{s_y}, \qquad z_{x_j} = \frac{x_j - \bar{x}_j}{s_{x_j}}.$$

A regression in standardized variables has the form

$$z_y = \beta_1^* z_{x_1} + \cdots + \beta_k^* z_{x_k} + u^*.$$

The standardized coefficient satisfies

$$\boxed{\beta_j^* = \hat{\beta}_j \frac{s_{x_j}}{s_y}}.$$

It measures the change in y, measured in standard deviations, associated with a one-standard-deviation change in x_j, holding the other regressors fixed.

5.2 Logarithmic Transformations

Logarithmic transformations are frequently used when proportional or percentage changes are more meaningful than absolute changes.
Recall the approximation

$$\Delta \log(x) \approx \frac{\Delta x}{x}$$

for relatively small changes.
Thus

$$100\Delta \log(x)$$

is approximately the percentage change in x.
Even for moderately large changes, logarithmic transformations often provide a useful representation of proportional variation.

5.3 Quadratic Regression Models

A linear regression model may include nonlinear functions of the explanatory variables.
Consider

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u.$$

Although the relationship between y and x is nonlinear, the model remains linear in the parameters $\beta_0, \beta_1, \beta_2$.
The partial effect of x is

$$\boxed{\frac{\partial \mathbb{E}(y \mid x)}{\partial x} = \beta_1 + 2\beta_2 x}.$$

Thus the effect of x depends on the value of x.
If $\beta_2 < 0$, the regression function is concave and the turning point is

$$\boxed{x^* = -\frac{\beta_1}{2\beta_2}}.$$

If $\beta_2 > 0$, the function is convex and the same expression gives the minimum point.

Example 5.3.1 (Experience and Wage)**.** A common specification is

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{exper} + \beta_2 \text{exper}^2 + u.$$

If

$$\beta_1 > 0, \qquad \beta_2 < 0,$$

then experience initially increases predicted wage, but the marginal effect declines as experience increases.
The marginal effect is

$$\frac{\partial \log(\text{wage})}{\partial \text{exper}} = \beta_1 + 2\beta_2 \text{exper}.$$

A mathematically valid turning point should also be checked against the range of observed data. An estimated extremum far outside the sample range may have little practical interpretation.

5.4 Interaction Terms

Interaction terms allow the effect of one explanatory variable to depend on the level of another.
Consider

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u.$$

Then

$$\boxed{\frac{\partial \mathbb{E}(y \mid x_1, x_2)}{\partial x_1} = \beta_1 + \beta_3 x_2}.$$

Similarly,

$$\frac{\partial \mathbb{E}(y \mid x_1, x_2)}{\partial x_2} = \beta_2 + \beta_3 x_1.$$

Thus the coefficient on x_1 alone is not the complete effect of x_1 once an interaction term is present.

Example 5.4.1 (Education and Experience)**.** Suppose

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 (\text{educ} \times \text{exper}) + u.$$

Then the effect of an additional year of education is

$$\frac{\partial \log(\text{wage})}{\partial \text{educ}} = \beta_1 + \beta_3 \text{exper}.$$

Hence the return to education is allowed to vary with work experience.

5.5 Average Partial Effects

When the marginal effect depends on the explanatory variables, a single coefficient may no longer summarize the effect of interest.
Suppose

$$\mathbb{E}(y \mid \boldsymbol{x}) = f(\boldsymbol{x}, \boldsymbol{\beta}).$$

The partial effect of x_j is

$$\frac{\partial f(\boldsymbol{x}, \boldsymbol{\beta})}{\partial x_j}.$$

The *Average Partial Effect* is the sample average of these individual partial effects:

$$\boxed{\text{APE}_j = \frac{1}{n} \sum_{i=1}^{n} \frac{\partial f(\boldsymbol{x}_i, \hat{\boldsymbol{\beta}})}{\partial x_j}}.$$

For the quadratic model

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u,$$

the estimated APE is

$$\widehat{\text{APE}} = \frac{1}{n} \sum_{i=1}^{n} \left(\hat{\beta}_1 + 2\hat{\beta}_2 x_i \right).$$

5.6 Reparametrization

Sometimes a hypothesis becomes easier to test after rewriting the model using a new set of parameters.
Suppose

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

and the hypothesis of interest is

$$H_0 : \beta_1 = \beta_2.$$

Define

$$\theta = \beta_1 - \beta_2.$$

Since

$$\beta_1 = \theta + \beta_2,$$

the model can be rewritten as

$$\begin{aligned} y &= \beta_0 + (\theta + \beta_2) x_1 + \beta_2 x_2 + u \\ &= \beta_0 + \theta x_1 + \beta_2 (x_1 + x_2) + u. \end{aligned}$$

The original hypothesis is now simply

$$H_0 : \theta = 0,$$

which can be tested using an ordinary t-test.

5.7 Adjusted R^2

The ordinary coefficient of determination is

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}}.$$

Adding explanatory variables cannot increase the residual sum of squares. Therefore R^2 never decreases when additional regressors are included, even when those regressors provide little useful information.
The adjusted coefficient of determination introduces a degrees-of-freedom correction:

$$\boxed{\bar{R}^2 = 1 - \frac{\text{SSR}/(n-k-1)}{\text{SST}/(n-1)}}.$$

Equivalently,

$$\bar{R}^2 = 1 - \frac{n-1}{n-k-1}(1-R^2).$$

Unlike ordinary R^2, adjusted R^2 may decrease when a new variable is added.
It may even be negative when the regression performs poorly relative to a model containing only an intercept.
Adjusted R^2 can be useful for comparing models with different numbers of regressors, but it should not be treated as a complete model selection criterion.

5.8 Dummy Variables

Qualitative information can be incorporated into regression models using dummy variables.

Definition 5.8.1 (Dummy Variable)**.** A dummy variable takes the values 0 and 1, indicating whether an observation belongs to a particular category.

For example, define

$$\text{female} = \begin{cases} 1, & \text{female}, \\ 0, & \text{male}. \end{cases}$$

Consider

$$\text{wage} = \beta_0 + \beta_1 \text{female} + u.$$

For males,

$$\mathbb{E}(\text{wage} \mid \text{female} = 0) = \beta_0.$$

For females,

$$\mathbb{E}(\text{wage} \mid \text{female} = 1) = \beta_0 + \beta_1.$$

Therefore,

$$\boxed{\beta_1 = \mathbb{E}(\text{wage} \mid \text{female} = 1) - \mathbb{E}(\text{wage} \mid \text{female} = 0)}.$$

Thus a dummy-variable coefficient represents an intercept shift between two groups.

5.9 Base Categories and the Dummy Variable Trap

Suppose a qualitative variable has G mutually exclusive categories. If the regression contains an intercept, only $G-1$ dummy variables should be included.
For example, with two categories,

$$D_1 + D_2 = 1.$$

Including both D_1 and D_2 together with an intercept produces perfect collinearity.
Therefore one category must be omitted. The omitted category is called the *base category* or *reference group*.
Suppose three education categories are represented by

$$D_1, \quad D_2, \quad D_3.$$

Using D_1 as the base group gives

$$y = \beta_0 + \beta_2 D_2 + \beta_3 D_3 + u.$$

Then β_2 measures the difference between groups 2 and 1, while β_3 measures the difference between groups 3 and 1.

5.10 Dummy Variables in Logarithmic Models

Consider

$$\log(y) = \beta_0 + \beta_1 D + u,$$

where D is a dummy variable.
The approximate percentage difference between the groups is

$$100\beta_1\%.$$

The exact percentage difference is

$$\boxed{100\left(e^{\beta_1} - 1\right)\%}.$$

The approximation is accurate when $|\beta_1|$ is small.

5.11 Interactions with Dummy Variables

Dummy variables can interact with quantitative explanatory variables.
Consider

$$y = \beta_0 + \beta_1 x + \delta_0 D + \delta_1 (Dx) + u.$$

For $D = 0$,

$$\mathbb{E}(y \mid x, D = 0) = \beta_0 + \beta_1 x.$$

For $D = 1$,

$$\mathbb{E}(y \mid x, D = 1) = (\beta_0 + \delta_0) + (\beta_1 + \delta_1)x.$$

Thus δ_0 represents an intercept difference and δ_1 represents a slope difference.

Example 5.11.1 (Different Wage Equations Across Groups)**.** Suppose

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \delta_0 D + \delta_1 (D \times \text{educ}) + u.$$

For the base group,

$$\frac{\partial \log(\text{wage})}{\partial \text{educ}} = \beta_1.$$

For the group with $D = 1$,

$$\frac{\partial \log(\text{wage})}{\partial \text{educ}} = \beta_1 + \delta_1.$$

Therefore δ_1 measures the difference in the return to education between the two groups.

5.12 The Chow Test

A stronger question is whether two groups have entirely different regression equations. Suppose the model contains k slope coefficients and an intercept. The null hypothesis is that every coefficient is identical across two groups:

$$H_0 : \boldsymbol{\beta}^{(1)} = \boldsymbol{\beta}^{(2)}.$$

Estimate the pooled regression under the restriction and obtain

$$\text{SSR}_r.$$

Then estimate the regression separately for the two groups and obtain

$$\text{SSR}_1 \qquad \text{and} \qquad \text{SSR}_2.$$

The unrestricted residual sum of squares is

$$\text{SSR}_{ur} = \text{SSR}_1 + \text{SSR}_2.$$

If there are $k + 1$ restrictions, the Chow statistic is

$$\boxed{F = \frac{(\text{SSR}_r - \text{SSR}_{ur})/(k+1)}{\text{SSR}_{ur}/[n_1 + n_2 - 2(k+1)]}}.$$

Rejecting the null provides evidence that at least one regression coefficient differs between the two groups.

5.13 The Linear Probability Model

A binary dependent variable can also be modeled using OLS.
Suppose

$$y \in \{0, 1\}.$$

Then

$$\mathbb{E}(y \mid X) = \mathbb{P}(y = 1 \mid X).$$

The linear probability model is

$$\boxed{\mathbb{P}(y = 1 \mid X) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k}.$$

Thus

$$\beta_j$$

measures the change in the predicted probability associated with a one-unit change in x_j, holding the other regressors fixed.

Example 5.13.1 (Employment Probability)**.** Suppose

$$\mathbb{P}(\text{employed} = 1 \mid X) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper}.$$

If

$$\hat{\beta}_1 = 0.025,$$

then one additional year of education is associated with an estimated increase of 0.025, or 2.5 percentage points, in the probability of employment, holding experience fixed.

Because y is binary,

$$\text{Var}(y \mid X) = p(X)[1 - p(X)],$$

where

$$p(X) = \mathbb{P}(y = 1 \mid X).$$

Therefore the error variance is generally not constant. The linear probability model naturally produces heteroskedasticity, which motivates the robust inference methods considered in the next chapter.

5.14 Model Misspecification

A regression model is misspecified when its functional form or set of explanatory variables does not adequately represent the conditional mean of the dependent variable.
Possible sources include

- omitted relevant variables;
- unnecessary variables;
- incorrect functional form;
- missing nonlinear terms;
- missing interaction terms.

For example, if the true conditional mean is

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_1 x + \beta_2 x^2,$$

but the estimated model is

$$y = \alpha_0 + \alpha_1 x + u,$$

then the functional form is misspecified whenever $\beta_2 \neq 0$.

5.15 Ramsey's RESET Test

The RESET test provides a general diagnostic for functional-form misspecification.
Begin with the original model

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u.$$

Estimate the model and obtain fitted values

$$\hat{y}.$$

Then estimate an augmented regression such as

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \delta_1 \hat{y}^2 + \delta_2 \hat{y}^3 + v.$$

Test

$$H_0 : \delta_1 = \delta_2 = 0.$$

An F-test can be used for this joint hypothesis.
If the added terms are jointly significant, there is evidence that the original specification is inadequate.
RESET is primarily a diagnostic test: rejection indicates that some form of specification problem may exist, but does not by itself identify the exact source of the problem.

5.16 Nonnested Models

Two models are *nonnested* when neither can be obtained from the other simply by imposing restrictions on its parameters.
Suppose two competing specifications are

$$y = X_1\boldsymbol{\beta} + u$$

and

$$y = X_2\boldsymbol{\delta} + v.$$

Estimate both models and obtain fitted values

$$\hat{y}_1 \qquad \text{and} \qquad \hat{y}_2.$$

One diagnostic approach augments the first model with $\hat{y}_2$:

$$y = X_1\boldsymbol{\beta} + \theta_1\hat{y}_2 + e.$$

If θ_1 is statistically significant, this provides evidence against the first specification.
Similarly, augment the second model with $\hat{y}_1$:

$$y = X_2\boldsymbol{\delta} + \theta_2\hat{y}_1 + e.$$

The significance of θ_2 provides information about the second specification.
Such procedures should be interpreted as specification diagnostics rather than automatic rules for selecting a unique correct model.

5.17 Proxy Variables

An important variable may affect the dependent variable but be unobservable.
Suppose the desired population model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3^* + u,$$

where x_3^* is unobserved.
A proxy variable x_3 is an observable variable intended to capture information about x_3^*.
Suppose

$$x_3^* = \delta_0 + \delta_3 x_3 + v.$$

Substituting gives

$$\begin{aligned} y &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3(\delta_0 + \delta_3 x_3 + v) + u \\ &= (\beta_0 + \beta_3\delta_0) + \beta_1 x_1 + \beta_2 x_2 + \beta_3\delta_3 x_3 + (\beta_3 v + u). \end{aligned}$$

Under suitable assumptions concerning the proxy and the composite error term, regression using x_3 may provide a useful solution to the omitted-variable problem.
This approach is sometimes described as a *proxy-variable* or *plug-in* solution.

Example 5.17.1 (Proxy for Ability)**.** Suppose ability affects wage but is not directly observed. A test score may be used as a proxy:

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{score} + u.$$

The validity of this strategy depends on whether the test score captures relevant variation in ability and whether the remaining proxy error is appropriately unrelated to the included explanatory variables.

5.18 Lagged Variables as Proxies

In some applications, a lagged value of a variable may serve as a proxy for an unobserved current characteristic.
The basic logic is the same as for other proxy variables: the lagged variable is useful when it contains information about the omitted factor while satisfying the necessary exogeneity conditions.
Whether such a proxy is valid must be justified from the structure of the particular model.

5.19 Random Coefficient Models

Regression coefficients may themselves vary across observational units.
Consider

$$y_i = (\alpha + c_i) + (\beta + d_i)x_i.$$

This can be rewritten as

$$y_i = \alpha + \beta x_i + (c_i + d_i x_i).$$

Thus the composite error is

$$u_i = c_i + d_i x_i.$$

If

$$\mathbb{E}(c_i \mid x_i) = 0, \qquad \mathbb{E}(d_i \mid x_i) = 0,$$

then

$$\mathbb{E}(u_i \mid x_i) = 0.$$

However, its conditional variance generally depends on x_i. For example, if c_i and d_i are uncorrelated,

$$\boxed{\text{Var}(u_i \mid x_i) = \sigma_c^2 + x_i^2 \sigma_d^2}.$$

Thus random coefficient models naturally generate heteroskedasticity.

5.20 Measurement Error in the Dependent Variable

Suppose the true dependent variable is y^*, but the observed variable contains measurement error:

$$y = y^* + e.$$

Suppose

$$y^* = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u.$$

Then the observed regression becomes

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + (u + e).$$

If the measurement error satisfies

$$\mathbb{E}(e \mid X) = 0,$$

it simply becomes part of the regression error.
In this classical case, OLS remains unbiased and consistent, although the additional noise may increase the variance of the estimator.

5.21 Measurement Error in an Explanatory Variable

Measurement error in an explanatory variable is more problematic.
Suppose the true model is

$$y = \beta_0 + \beta_1 x^* + u,$$

but instead of observing x^*, we observe

$$x = x^* + e.$$

Since

$$x^* = x - e,$$

substitution gives

$$\begin{aligned} y &= \beta_0 + \beta_1(x - e) + u \\ &= \beta_0 + \beta_1 x + (u - \beta_1 e). \end{aligned}$$

The new regression error is therefore

$$v = u - \beta_1 e.$$

But the observed explanatory variable contains e, so

$$\operatorname{Cov}(x, v)$$

is generally nonzero.
Under the classical errors-in-variables assumptions,

$$\operatorname{Cov}(x^*, e) = 0, \qquad \operatorname{Cov}(u, e) = 0,$$

we obtain

$$\begin{aligned} \operatorname{Cov}(x, v) &= \operatorname{Cov}(x^* + e, u - \beta_1 e) \\ &= -\beta_1 \operatorname{Var}(e). \end{aligned}$$

Thus the observed regressor is endogenous and OLS is generally biased and inconsistent.

5.22 Attenuation Bias

Under the classical measurement-error model,

$$y = \beta_0 + \beta_1 x^* + u, \qquad x = x^* + e,$$

the probability limit of the OLS slope is

$$\boxed{\operatorname{plim} \hat{\beta}_1 = \beta_1 \frac{\operatorname{Var}(x^*)}{\operatorname{Var}(x^*) + \operatorname{Var}(e)}}.$$

Define

$$\lambda = \frac{\operatorname{Var}(x^*)}{\operatorname{Var}(x^*) + \operatorname{Var}(e)}.$$

Then $0 < \lambda < 1$, so

$$\operatorname{plim} \hat{\beta}_1 = \lambda \beta_1.$$

The coefficient is therefore biased toward zero. This phenomenon is called *attenuation bias*.

5.23 Missing Data

Missing observations may occur in either explanatory variables or the dependent variable. If observations are missing completely independently of the regression error and explanatory variables, using the available complete observations mainly reduces the effective sample size.
If the probability that an observation is missing depends systematically on variables related to the regression error, the resulting sample may no longer be representative of the population.
In such cases, deleting incomplete observations can introduce selection problems and affect the validity of OLS.
The consequences therefore depend on the mechanism generating the missing data rather than simply on the presence of missing observations.

5.24 Outliers and Influential Observations

An outlier is an observation whose value is unusually far from the general pattern of the data.
Because OLS minimizes squared residuals,

$$\sum_{i=1}^{n} \hat{u}_i^2,$$

large residuals receive disproportionately large weight.
An observation may be unusual because of

- an extreme value of the dependent variable;
- an extreme value of an explanatory variable;
- measurement or recording error;
- genuine but rare behavior.

An observation is particularly important when its inclusion or removal substantially changes the estimated regression coefficients. Such an observation is called *influential*.
Outliers should therefore be investigated rather than automatically deleted. Their economic or data-generating interpretation matters for deciding whether they represent errors or genuine observations.

5.25 Summary of Model Specification

Regression design involves more than selecting a set of explanatory variables. The functional form determines how coefficients should be interpreted and which partial effects are implied by the model.
Logarithms describe proportional changes, quadratic terms allow marginal effects to vary with the level of a variable, and interaction terms allow effects to depend on other characteristics. Dummy variables incorporate qualitative information and allow regression equations to differ across groups.
Specification diagnostics such as RESET can indicate whether a proposed model is inadequate, while proxy variables may partially address unobserved explanatory factors. Measurement error, missing data, and outliers illustrate that the quality of the observed variables is as important as the algebra of OLS itself.
Several of these issues also affect the variance of the regression error. In particular, binary dependent variables, random coefficients, and many other specifications naturally

produce nonconstant conditional variance. The next chapter studies heteroskedasticity and the methods used to conduct valid estimation and inference when homoskedasticity fails.

Chapter 6

Heteroskedasticity and Weighted Least Squares

6.1 Homoskedasticity and Heteroskedasticity

In the classical linear regression model, homoskedasticity requires that the conditional variance of the error term remain constant across observations:

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2.$$

When this assumption fails, the regression errors are *heteroskedastic*.

Definition 6.1.1 (Heteroskedasticity)**.** A regression model is heteroskedastic if

$$\mathrm{Var}(u_i \mid X_i)$$

depends on the explanatory variables or differs systematically across observations.

A general representation is

$$\boxed{\mathrm{Var}(u_i \mid X_i) = \sigma_i^2},$$

where σ_i^2 is not constant across i.
Equivalently, the conditional covariance matrix of the error vector is

$$\mathrm{Var}(\boldsymbol{u} \mid X) = \Omega,$$

where

$$\Omega = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix}.$$

Under homoskedasticity,

$$\Omega = \sigma^2 I_n.$$

Under heteroskedasticity, the diagonal elements of Ω are not all equal.

6.2 Why Heteroskedasticity Arises

Heteroskedasticity is common in cross-sectional data because the variation of an outcome often changes with the scale or characteristics of the observational unit.

Example 6.2.1 (Income and Education)**.** Consider

$$\mathrm{income}_i = \beta_0 + \beta_1 \mathrm{educ}_i + u_i.$$

Among individuals with relatively little education, incomes may be concentrated within a relatively narrow range. Among highly educated individuals, however, observed incomes may range from ordinary salaries to extremely high professional or managerial incomes. Thus it is plausible that

$$\mathrm{Var}(u_i \mid \mathrm{educ}_i)$$

increases with education, even if the conditional mean equation is correctly specified.

Heteroskedasticity concerns the conditional variance rather than the conditional mean. Therefore a regression may have

$$\mathbb{E}(u_i \mid X_i) = 0$$

while simultaneously satisfying

$$\mathrm{Var}(u_i \mid X_i) \neq \sigma^2.$$

6.3 OLS under Heteroskedasticity

Consider

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{u}.$$

The OLS estimator is

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \boldsymbol{y}$$

and therefore

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{u}.$$

If

$$\mathbb{E}(\boldsymbol{u} \mid X) = 0,$$

then

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta}.$$

Thus heteroskedasticity by itself does not destroy the unbiasedness of OLS.
However, the variance changes. In general,

$$\begin{aligned} \operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) &= (X^\top X)^{-1} X^\top \operatorname{Var}(\boldsymbol{u} \mid X) X (X^\top X)^{-1} \\ &= (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}. \end{aligned}$$

Therefore,

$$\boxed{\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}}.$$

Under homoskedasticity,

$$\Omega = \sigma^2 I_n,$$

and the familiar formula

$$\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}$$

is recovered.
Under heteroskedasticity, this formula is generally incorrect.

6.4 Consequences of Heteroskedasticity

The main consequences can be separated into estimation and inference.
If zero conditional mean still holds, then

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta},$$

so OLS remains unbiased.
Under suitable large-sample conditions, OLS can also remain consistent.
However,

$$\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) \neq \sigma^2 (X^\top X)^{-1}.$$

Consequently, standard errors computed under the homoskedasticity assumption are generally incorrect.
Since t- and F-statistics depend on estimated variances, incorrect standard errors may lead to invalid inference.
Furthermore, the Gauss–Markov efficiency result no longer applies in its usual form. OLS is not generally the minimum-variance linear unbiased estimator when the error variance differs systematically across observations.
Thus the central problem is not necessarily the point estimate $\hat{\boldsymbol{\beta}}$, but the variance associated with that estimate.

6.5 Detecting Heteroskedasticity

The basic idea behind heteroskedasticity tests is to examine whether the magnitude of the regression residuals is systematically related to observable explanatory variables. Since

$$\mathbb{E}(u_i^2 \mid X_i) = \mathrm{Var}(u_i \mid X_i)$$

when

$$\mathbb{E}(u_i \mid X_i) = 0,$$

the squared residual

$$\hat{u}_i^2$$

provides an observable approximation to the conditional error variance.
A typical auxiliary regression therefore uses squared residuals as the dependent variable.

6.6 The Breusch–Pagan Test

Suppose the original regression is

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i.$$

Under homoskedasticity,

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2.$$

Equivalently,

$$\mathbb{E}(u_i^2 \mid X_i) = \sigma^2.$$

The Breusch–Pagan test models the conditional variance as a linear function of the explanatory variables:

$$u_i^2 = \delta_0 + \delta_1 x_{i1} + \cdots + \delta_k x_{ik} + v_i.$$

Under homoskedasticity,

$$\boxed{H_0 : \delta_1 = \cdots = \delta_k = 0}.$$

The practical procedure is:

1. Estimate the original regression by OLS and obtain residuals $\hat{u}_i$.
2. Form the squared residuals $\hat{u}_i^2$.
3. Estimate the auxiliary regression

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \cdots + \delta_k x_{ik} + v_i.$$

4. Test whether the slope coefficients are jointly zero.

An F-test may be used:

$$H_0 : \delta_1 = \cdots = \delta_k = 0.$$

Alternatively, let R^2_{aux} denote the R^2 from the auxiliary regression. The LM statistic is

$$\boxed{LM = nR^2_{\text{aux}}}.$$

Under the null hypothesis,

$$\boxed{LM \xrightarrow{d} \chi^2_k}.$$

A large value of LM provides evidence against homoskedasticity.

6.7 The White Test

The Breusch–Pagan test specifies the conditional variance as a linear function of selected explanatory variables.
The White test is more flexible because it allows nonlinear variance patterns.
For example, with two explanatory variables, estimate the original regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$$

and obtain $\hat{u}_i$.
Then estimate the auxiliary regression

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \delta_2 x_{i2} + \delta_3 x_{i1}^2 + \delta_4 x_{i2}^2 + \delta_5 x_{i1} x_{i2} + v_i.$$

The null hypothesis is

$$H_0 : \delta_1 = \delta_2 = \cdots = \delta_5 = 0.$$

If the auxiliary regression contains q slope terms, the White LM statistic is

$$\boxed{LM = nR^2_{\text{aux}} \xrightarrow{d} \chi^2_q}.$$

The White test does not require specifying one particular functional form for the conditional variance.
However, including all squares and cross-products can create many auxiliary regressors when the original model contains many explanatory variables.

6.8 An Alternative White-Type Test

A simpler specification test can be constructed using fitted values.
After estimating the original regression, obtain

$$\hat{y}_i.$$

Then estimate

$$\hat{u}_i^2 = \delta_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + v_i.$$

Test

$$H_0 : \delta_1 = \delta_2 = 0.$$

The corresponding LM statistic is

$$\boxed{LM = nR^2_{\text{aux}}}.$$

Under the null,

$$LM \xrightarrow{d} \chi^2_2.$$

This version reduces the number of variables required in the auxiliary regression.

6.9 Weighted Least Squares

If the form of heteroskedasticity is known, the regression can be transformed so that the transformed error has constant variance.
Suppose

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

and

$$\boxed{\mathrm{Var}(u_i \mid X_i) = \sigma^2 h_i},$$

where $h_i > 0$ is known.
Divide the entire regression by $\sqrt{h_i}$:

$$\frac{y_i}{\sqrt{h_i}} = \beta_0 \frac{1}{\sqrt{h_i}} + \beta_1 \frac{x_{i1}}{\sqrt{h_i}} + \cdots + \beta_k \frac{x_{ik}}{\sqrt{h_i}} + \frac{u_i}{\sqrt{h_i}}.$$

Define

$$y_i^* = \frac{y_i}{\sqrt{h_i}}, \qquad x_{ij}^* = \frac{x_{ij}}{\sqrt{h_i}}, \qquad u_i^* = \frac{u_i}{\sqrt{h_i}}.$$

Then

$$y_i^* = \beta_0 \frac{1}{\sqrt{h_i}} + \beta_1 x_{i1}^* + \cdots + \beta_k x_{ik}^* + u_i^*.$$

The transformed error satisfies

$$\begin{aligned} \mathrm{Var}(u_i^* \mid X_i) &= \mathrm{Var}\left(\frac{u_i}{\sqrt{h_i}} \middle| X_i \right) \\ &= \frac{1}{h_i} \mathrm{Var}(u_i \mid X_i) \\ &= \sigma^2. \end{aligned}$$

Thus the transformed model is homoskedastic.
OLS applied to the transformed regression is called *Weighted Least Squares*.

6.10 The WLS Objective Function

OLS minimizes

$$\sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik} \right)^2.$$

After dividing by $\sqrt{h_i}$, WLS minimizes

$$\boxed{\sum_{i=1}^{n} \frac{\left(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik} \right)^2}{h_i}}.$$

Thus observation i receives weight

$$w_i = \frac{1}{h_i}.$$

Since

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2 h_i,$$

observations with larger conditional variance receive smaller weights.
Therefore WLS gives greater weight to more precisely measured observations and less weight to observations with greater error variance.

6.11 Matrix Form of WLS

Suppose

$$\mathrm{Var}(\boldsymbol{u} \mid X) = \sigma^2 \Omega,$$

where

$$\Omega = \mathrm{diag}(h_1, \ldots, h_n).$$

The WLS estimator can be written as

$$\boxed{\hat{\boldsymbol{\beta}}_{WLS} = (X^\top \Omega^{-1} X)^{-1} X^\top \Omega^{-1} \boldsymbol{y}}.$$

If the variance specification is correct, then

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}_{WLS} \mid X) = \sigma^2 (X^\top \Omega^{-1} X)^{-1}.$$

When $\Omega = I_n$, WLS reduces to ordinary least squares.

6.12 Feasible Weighted Least Squares

In practice, the variance function h_i is usually unknown.
A feasible weighted least squares procedure estimates the variance function first and then uses the estimated values as weights.
A common specification is

$$\mathrm{Var}(u_i \mid X_i) = \exp(\gamma_0 + \gamma_1 z_{i1} + \cdots + \gamma_m z_{im}).$$

The exponential form guarantees a positive variance.
The FWLS procedure can be organized as follows.

1. Estimate the original model

   $$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

   by OLS and obtain residuals $\hat{u}_i$.

2. Construct

   $$\log(\hat{u}_i^2).$$

3. Estimate the variance equation

   $$\log(\hat{u}_i^2) = \gamma_0 + \gamma_1 z_{i1} + \cdots + \gamma_m z_{im} + v_i.$$

4. Obtain fitted values

   $$\hat{g}_i = \hat{\gamma}_0 + \hat{\gamma}_1 z_{i1} + \cdots + \hat{\gamma}_m z_{im}.$$

5. Estimate

   $$\hat{h}_i = \exp(\hat{g}_i).$$

6. Use weights

   $$\boxed{w_i = \frac{1}{\hat{h}_i}}$$

   and estimate the original regression by WLS.

Because the true variance function has been replaced by an estimate, this procedure is called *Feasible Weighted Least Squares*.

6.13 WLS in the Linear Probability Model

The linear probability model provides a natural example of known heteroskedastic structure.

Suppose

$$y_i \in \{0, 1\}$$

and

$$p_i = \mathbb{P}(y_i = 1 \mid X_i).$$

Since y_i is Bernoulli conditional on X_i,

$$\mathbb{E}(y_i \mid X_i) = p_i$$

and

$$\boxed{\mathrm{Var}(y_i \mid X_i) = p_i(1 - p_i)}.$$

In the linear probability model,

$$p_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.$$

Therefore the regression error satisfies

$$\mathrm{Var}(u_i \mid X_i) = p_i(1 - p_i),$$

which is heteroskedastic unless p_i is constant.

Because the true probabilities are unknown, first estimate the linear probability model by OLS and obtain fitted probabilities $\hat{p}_i$.

Then estimate

$$\hat{h}_i = \hat{p}_i(1 - \hat{p}_i).$$

The corresponding weights are

$$\boxed{w_i = \frac{1}{\hat{p}_i(1 - \hat{p}_i)}}.$$

The transformed regression uses

$$\frac{1}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}}$$

as the observation-specific scaling factor.

This is an application of feasible weighted least squares because the conditional variances are estimated rather than known.

6.14 OLS versus WLS

The distinction between OLS and WLS can be summarized through their objective functions.

OLS solves

$$\hat{\boldsymbol{\beta}}_{OLS} = \arg\min_{\boldsymbol{b}} \sum_{i=1}^{n} e_i^2.$$

WLS solves

$$\hat{\boldsymbol{\beta}}_{WLS} = \arg\min_{\boldsymbol{b}} \sum_{i=1}^{n} w_i e_i^2.$$

Under homoskedasticity, all observations have the same error variance, so equal weighting is appropriate.
Under known heteroskedasticity,

$$w_i \propto \frac{1}{\mathrm{Var}(u_i \mid X_i)}.$$

Thus WLS compensates for unequal conditional variances by giving less weight to observations whose outcomes are intrinsically more variable.

6.15 Summary of Heteroskedasticity

Under heteroskedasticity,

$$\mathrm{Var}(u_i \mid X_i)$$

varies across observations.
If

$$\mathbb{E}(u_i \mid X_i) = 0,$$

OLS may remain unbiased and consistent, but its homoskedastic variance formula is no longer valid.
The correct conditional covariance matrix of OLS is

$$\boxed{\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}}.$$

Heteroskedasticity can be investigated through auxiliary regressions. The Breusch–Pagan and White tests use squared residuals and commonly produce statistics of the form

$$LM = nR^2.$$

If the conditional variance has the known structure

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2 h_i,$$

weighted least squares transforms the model using $1/\sqrt{h_i}$.
When h_i is unknown, feasible weighted least squares first estimates the variance function and then applies estimated weights.
Thus this chapter separates two related problems:

detecting nonconstant variance	and	using that variance structure in estimation.

The next chapter turns from cross-sectional regression to time series data, where observations are indexed by time and dependence across periods becomes an additional part of the regression structure.

Chapter 7

Heteroskedasticity and Weighted Least Squares

7.1 Homoskedasticity and Heteroskedasticity

In the classical linear regression model, homoskedasticity requires that the conditional variance of the error term remain constant across observations:

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2.$$

When this assumption fails, the regression errors are *heteroskedastic*.

Definition 7.1.1 (Heteroskedasticity)**.** A regression model is heteroskedastic if

$$\mathrm{Var}(u_i \mid X_i)$$

depends on the explanatory variables or differs systematically across observations.

A general representation is

$$\boxed{\mathrm{Var}(u_i \mid X_i) = \sigma_i^2},$$

where σ_i^2 is not constant across i.
Equivalently, the conditional covariance matrix of the error vector is

$$\mathrm{Var}(\boldsymbol{u} \mid X) = \Omega,$$

where

$$\Omega = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix}.$$

Under homoskedasticity,

$$\Omega = \sigma^2 I_n.$$

Under heteroskedasticity, the diagonal elements of Ω are not all equal.

7.2 Why Heteroskedasticity Arises

Heteroskedasticity is common in cross-sectional data because the variation of an outcome often changes with the scale or characteristics of the observational unit.

Example 7.2.1 (Income and Education)**.** Consider

$$\mathrm{income}_i = \beta_0 + \beta_1 \mathrm{educ}_i + u_i.$$

Among individuals with relatively little education, incomes may be concentrated within a relatively narrow range. Among highly educated individuals, however, observed incomes may range from ordinary salaries to extremely high professional or managerial incomes. Thus it is plausible that

$$\mathrm{Var}(u_i \mid \mathrm{educ}_i)$$

increases with education, even if the conditional mean equation is correctly specified.

Heteroskedasticity concerns the conditional variance rather than the conditional mean. Therefore a regression may have

$$\mathbb{E}(u_i \mid X_i) = 0$$

while simultaneously satisfying

$$\mathrm{Var}(u_i \mid X_i) \neq \sigma^2.$$

7.3 OLS under Heteroskedasticity

Consider

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{u}.$$

The OLS estimator is

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \boldsymbol{y}$$

and therefore

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{u}.$$

If

$$\mathbb{E}(\boldsymbol{u} \mid X) = 0,$$

then

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta}.$$

Thus heteroskedasticity by itself does not destroy the unbiasedness of OLS.
However, the variance changes. In general,

$$\begin{aligned} \operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) &= (X^\top X)^{-1} X^\top \operatorname{Var}(\boldsymbol{u} \mid X) X (X^\top X)^{-1} \\ &= (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}. \end{aligned}$$

Therefore,

$$\boxed{\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}}.$$

Under homoskedasticity,

$$\Omega = \sigma^2 I_n,$$

and the familiar formula

$$\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}$$

is recovered.
Under heteroskedasticity, this formula is generally incorrect.

7.4 Consequences of Heteroskedasticity

The main consequences can be separated into estimation and inference.
If zero conditional mean still holds, then

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta},$$

so OLS remains unbiased.
Under suitable large-sample conditions, OLS can also remain consistent.
However,

$$\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid X) \neq \sigma^2 (X^\top X)^{-1}.$$

Consequently, standard errors computed under the homoskedasticity assumption are generally incorrect.
Since t- and F-statistics depend on estimated variances, incorrect standard errors may lead to invalid inference.
Furthermore, the Gauss–Markov efficiency result no longer applies in its usual form. OLS is not generally the minimum-variance linear unbiased estimator when the error variance differs systematically across observations.
Thus the central problem is not necessarily the point estimate $\hat{\boldsymbol{\beta}}$, but the variance associated with that estimate.

7.5 Detecting Heteroskedasticity

The basic idea behind heteroskedasticity tests is to examine whether the magnitude of the regression residuals is systematically related to observable explanatory variables. Since

$$\mathbb{E}(u_i^2 \mid X_i) = \mathrm{Var}(u_i \mid X_i)$$

when

$$\mathbb{E}(u_i \mid X_i) = 0,$$

the squared residual

$$\hat{u}_i^2$$

provides an observable approximation to the conditional error variance.
A typical auxiliary regression therefore uses squared residuals as the dependent variable.

7.6 The Breusch–Pagan Test

Suppose the original regression is

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i.$$

Under homoskedasticity,

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2.$$

Equivalently,

$$\mathbb{E}(u_i^2 \mid X_i) = \sigma^2.$$

The Breusch–Pagan test models the conditional variance as a linear function of the explanatory variables:

$$u_i^2 = \delta_0 + \delta_1 x_{i1} + \cdots + \delta_k x_{ik} + v_i.$$

Under homoskedasticity,

$$\boxed{H_0 : \delta_1 = \cdots = \delta_k = 0}.$$

The practical procedure is:

1. Estimate the original regression by OLS and obtain residuals $\hat{u}_i$.
2. Form the squared residuals $\hat{u}_i^2$.
3. Estimate the auxiliary regression

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \cdots + \delta_k x_{ik} + v_i.$$

4. Test whether the slope coefficients are jointly zero.

An F-test may be used:

$$H_0 : \delta_1 = \cdots = \delta_k = 0.$$

Alternatively, let R^2_{aux} denote the R^2 from the auxiliary regression. The LM statistic is

$$\boxed{LM = nR^2_{\text{aux}}}.$$

Under the null hypothesis,

$$\boxed{LM \xrightarrow{d} \chi^2_k}.$$

A large value of LM provides evidence against homoskedasticity.

7.7 The White Test

The Breusch–Pagan test specifies the conditional variance as a linear function of selected explanatory variables.
The White test is more flexible because it allows nonlinear variance patterns.
For example, with two explanatory variables, estimate the original regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$$

and obtain $\hat{u}_i$.
Then estimate the auxiliary regression

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \delta_2 x_{i2} + \delta_3 x_{i1}^2 + \delta_4 x_{i2}^2 + \delta_5 x_{i1} x_{i2} + v_i.$$

The null hypothesis is

$$H_0 : \delta_1 = \delta_2 = \cdots = \delta_5 = 0.$$

If the auxiliary regression contains q slope terms, the White LM statistic is

$$\boxed{LM = nR^2_{\text{aux}} \xrightarrow{d} \chi^2_q}.$$

The White test does not require specifying one particular functional form for the conditional variance.
However, including all squares and cross-products can create many auxiliary regressors when the original model contains many explanatory variables.

7.8 An Alternative White-Type Test

A simpler specification test can be constructed using fitted values.
After estimating the original regression, obtain

$$\hat{y}_i.$$

Then estimate

$$\hat{u}_i^2 = \delta_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + v_i.$$

Test

$$H_0 : \delta_1 = \delta_2 = 0.$$

The corresponding LM statistic is

$$\boxed{LM = nR^2_{\text{aux}}}.$$

Under the null,

$$LM \xrightarrow{d} \chi^2_2.$$

This version reduces the number of variables required in the auxiliary regression.

7.9 Weighted Least Squares

If the form of heteroskedasticity is known, the regression can be transformed so that the transformed error has constant variance.
Suppose

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

and

$$\boxed{\mathrm{Var}(u_i \mid X_i) = \sigma^2 h_i},$$

where $h_i > 0$ is known.
Divide the entire regression by $\sqrt{h_i}$:

$$\frac{y_i}{\sqrt{h_i}} = \beta_0 \frac{1}{\sqrt{h_i}} + \beta_1 \frac{x_{i1}}{\sqrt{h_i}} + \cdots + \beta_k \frac{x_{ik}}{\sqrt{h_i}} + \frac{u_i}{\sqrt{h_i}}.$$

Define

$$y_i^* = \frac{y_i}{\sqrt{h_i}}, \qquad x_{ij}^* = \frac{x_{ij}}{\sqrt{h_i}}, \qquad u_i^* = \frac{u_i}{\sqrt{h_i}}.$$

Then

$$y_i^* = \beta_0 \frac{1}{\sqrt{h_i}} + \beta_1 x_{i1}^* + \cdots + \beta_k x_{ik}^* + u_i^*.$$

The transformed error satisfies

$$\begin{aligned} \mathrm{Var}(u_i^* \mid X_i) &= \mathrm{Var}\left(\frac{u_i}{\sqrt{h_i}} \middle| X_i \right) \\ &= \frac{1}{h_i} \mathrm{Var}(u_i \mid X_i) \\ &= \sigma^2. \end{aligned}$$

Thus the transformed model is homoskedastic.
OLS applied to the transformed regression is called *Weighted Least Squares*.

7.10 The WLS Objective Function

OLS minimizes

$$\sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik} \right)^2.$$

After dividing by $\sqrt{h_i}$, WLS minimizes

$$\boxed{\sum_{i=1}^{n} \frac{\left(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik} \right)^2}{h_i}}.$$

Thus observation i receives weight

$$w_i = \frac{1}{h_i}.$$

Since

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2 h_i,$$

observations with larger conditional variance receive smaller weights.
Therefore WLS gives greater weight to more precisely measured observations and less weight to observations with greater error variance.

7.11 Matrix Form of WLS

Suppose

$$\mathrm{Var}(\boldsymbol{u} \mid X) = \sigma^2 \Omega,$$

where

$$\Omega = \mathrm{diag}(h_1, \ldots, h_n).$$

The WLS estimator can be written as

$$\boxed{\hat{\boldsymbol{\beta}}_{WLS} = (X^\top \Omega^{-1} X)^{-1} X^\top \Omega^{-1} \boldsymbol{y}}.$$

If the variance specification is correct, then

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}_{WLS} \mid X) = \sigma^2 (X^\top \Omega^{-1} X)^{-1}.$$

When $\Omega = I_n$, WLS reduces to ordinary least squares.

7.12 Feasible Weighted Least Squares

In practice, the variance function h_i is usually unknown.
A feasible weighted least squares procedure estimates the variance function first and then uses the estimated values as weights.
A common specification is

$$\mathrm{Var}(u_i \mid X_i) = \exp(\gamma_0 + \gamma_1 z_{i1} + \cdots + \gamma_m z_{im}).$$

The exponential form guarantees a positive variance.
The FWLS procedure can be organized as follows.

1. Estimate the original model

   $$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

   by OLS and obtain residuals $\hat{u}_i$.

2. Construct

   $$\log(\hat{u}_i^2).$$

3. Estimate the variance equation

   $$\log(\hat{u}_i^2) = \gamma_0 + \gamma_1 z_{i1} + \cdots + \gamma_m z_{im} + v_i.$$

4. Obtain fitted values

   $$\hat{g}_i = \hat{\gamma}_0 + \hat{\gamma}_1 z_{i1} + \cdots + \hat{\gamma}_m z_{im}.$$

5. Estimate

   $$\hat{h}_i = \exp(\hat{g}_i).$$

6. Use weights

   $$\boxed{w_i = \frac{1}{\hat{h}_i}}$$

   and estimate the original regression by WLS.

Because the true variance function has been replaced by an estimate, this procedure is called *Feasible Weighted Least Squares*.

7.13 WLS in the Linear Probability Model

The linear probability model provides a natural example of known heteroskedastic structure.

Suppose

$$y_i \in \{0, 1\}$$

and

$$p_i = \mathbb{P}(y_i = 1 \mid X_i).$$

Since y_i is Bernoulli conditional on X_i,

$$\mathbb{E}(y_i \mid X_i) = p_i$$

and

$$\boxed{\mathrm{Var}(y_i \mid X_i) = p_i(1 - p_i)}.$$

In the linear probability model,

$$p_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.$$

Therefore the regression error satisfies

$$\mathrm{Var}(u_i \mid X_i) = p_i(1 - p_i),$$

which is heteroskedastic unless p_i is constant.

Because the true probabilities are unknown, first estimate the linear probability model by OLS and obtain fitted probabilities $\hat{p}_i$.

Then estimate

$$\hat{h}_i = \hat{p}_i(1 - \hat{p}_i).$$

The corresponding weights are

$$\boxed{w_i = \frac{1}{\hat{p}_i(1 - \hat{p}_i)}}.$$

The transformed regression uses

$$\frac{1}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}}$$

as the observation-specific scaling factor.

This is an application of feasible weighted least squares because the conditional variances are estimated rather than known.

7.14 OLS versus WLS

The distinction between OLS and WLS can be summarized through their objective functions.

OLS solves

$$\hat{\boldsymbol{\beta}}_{OLS} = \arg\min_{\boldsymbol{b}} \sum_{i=1}^{n} e_i^2.$$

WLS solves

$$\hat{\boldsymbol{\beta}}_{WLS} = \arg\min_{\boldsymbol{b}} \sum_{i=1}^{n} w_i e_i^2.$$

Under homoskedasticity, all observations have the same error variance, so equal weighting is appropriate.
Under known heteroskedasticity,

$$w_i \propto \frac{1}{\mathrm{Var}(u_i \mid X_i)}.$$

Thus WLS compensates for unequal conditional variances by giving less weight to observations whose outcomes are intrinsically more variable.

7.15 Summary of Heteroskedasticity

Under heteroskedasticity,

$$\mathrm{Var}(u_i \mid X_i)$$

varies across observations.
If

$$\mathbb{E}(u_i \mid X_i) = 0,$$

OLS may remain unbiased and consistent, but its homoskedastic variance formula is no longer valid.
The correct conditional covariance matrix of OLS is

$$\boxed{\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}}.$$

Heteroskedasticity can be investigated through auxiliary regressions. The Breusch–Pagan and White tests use squared residuals and commonly produce statistics of the form

$$LM = nR^2.$$

If the conditional variance has the known structure

$$\mathrm{Var}(u_i \mid X_i) = \sigma^2 h_i,$$

weighted least squares transforms the model using $1/\sqrt{h_i}$.
When h_i is unknown, feasible weighted least squares first estimates the variance function and then applies estimated weights.
Thus this chapter separates two related problems:

detecting nonconstant variance and using that variance structure in estimation.

The next chapter turns from cross-sectional regression to time series data, where observations are indexed by time and dependence across periods becomes an additional part of the regression structure.

Chapter 8

Regression with Time Series Data

8.1 The Nature of Time Series Data

Time series data consist of observations indexed by time:

$$\{y_t, x_{t1}, \ldots, x_{tk}\}_{t=1}^{T}.$$

Unlike a random cross-sectional sample, the ordering of observations is essential. The observation at time t may depend on variables or outcomes from previous periods.
Examples include GDP, inflation, interest rates, unemployment, stock prices, consumption, investment, and aggregate production.
A general time series regression can be written as

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_k x_{tk} + u_t.$$

The algebra of OLS is unchanged, but the statistical assumptions require additional attention because observations may be dependent across time.

8.2 Static Time Series Models

The simplest time series regression is a static model.

Model 8.2.1 (Static Model)**.** A static time series model has the form

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_k x_{tk} + u_t.$$

Only current values of the explanatory variables appear on the right-hand side.
Thus the effect of x_{tj} on y_t is assumed to occur within the same period.

Example 8.2.2 (Consumption and Income)**.** A simple aggregate consumption model is

$$C_t = \beta_0 + \beta_1 Y_t + u_t,$$

where C_t is consumption and Y_t is disposable income.
The coefficient β_1 measures the contemporaneous change in consumption associated with a one-unit change in income.

8.3 Finite Distributed Lag Models

Economic responses are often delayed. A change in an explanatory variable may affect the dependent variable over several periods.

Model 8.3.1 (Finite Distributed Lag Model)**.** A finite distributed lag model is

$$\boxed{y_t = \alpha_0 + \delta_0 x_t + \delta_1 x_{t-1} + \cdots + \delta_q x_{t-q} + u_t}.$$

The current value x_t and a finite number of its lagged values enter the regression.
The coefficient δ_j measures the effect at horizon j of a change in x, provided the relevant interpretation is justified.

8.4 Impact and Lag Effects

Suppose x_t increases permanently by one unit beginning at time t.
At time t, the immediate effect on y_t is

$$\boxed{\delta_0}.$$

This is called the *impact propensity* or *impact multiplier*.
One period later, the additional effect is

$$\delta_1,$$

and after j periods, the additional effect is

$$\delta_j.$$

Thus the coefficients

$$\delta_0, \delta_1, \ldots, \delta_q$$

describe the dynamic response of y over time.

8.5 Long-Run Propensity

If the increase in x_t is permanent, then after all lagged effects have occurred, the total change in y is

$$\delta_0 + \delta_1 + \cdots + \delta_q.$$

Definition 8.5.1 (Long-Run Propensity)**.** For the distributed lag model

$$y_t = \alpha_0 + \sum_{j=0}^{q} \delta_j x_{t-j} + u_t,$$

the long-run propensity is

$$\boxed{\text{LRP} = \sum_{j=0}^{q} \delta_j}.$$

It is also called the *long-run multiplier* or *cumulative effect*.

Example 8.5.2 (Advertising and Sales)**.** Suppose

$$\text{sales}_t = \alpha_0 + \delta_0 \text{advert}_t + \delta_1 \text{advert}_{t-1} + \delta_2 \text{advert}_{t-2} + u_t.$$

Then δ_0 measures the immediate effect of additional advertising, while

$$\delta_0 + \delta_1 + \delta_2$$

measures its total long-run effect if the increase in advertising is permanent.

8.6 OLS Assumptions for Time Series Regression

Many assumptions from cross-sectional regression remain relevant, but their time-series interpretation is different.
A basic set of assumptions includes:

Assumption 8.6.1 (Linear in Parameters)**.**

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_k x_{tk} + u_t.$$

Assumption 8.6.2 (No Perfect Collinearity)**.** No explanatory variable is an exact linear combination of the others over the observed time periods.

Assumption 8.6.3 (Zero Conditional Mean)**.** A strong time-series exogeneity condition is

$$\boxed{\mathbb{E}(u_t \mid X_1, \ldots, X_T) = 0}.$$

This condition requires the current error term to be uncorrelated not only with current regressors, but also with past and future values of the explanatory variables.
Under suitable versions of these assumptions, OLS remains unbiased.

8.7 Homoskedasticity and No Serial Correlation

The classical time series regression additionally assumes

$$\mathrm{Var}(u_t \mid X) = \sigma^2$$

for every t.
It also assumes that errors from different time periods are uncorrelated:

$$\boxed{\mathrm{Cov}(u_t, u_s \mid X) = 0, \qquad t \neq s}.$$

This is the *no serial correlation* assumption.
Under homoskedasticity and no serial correlation,

$$\mathrm{Var}(\boldsymbol{u} \mid X) = \sigma^2 I_T.$$

Then the familiar OLS variance formula applies:

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}.$$

If the errors are also normally distributed, exact finite-sample t- and F-inference can be obtained as in cross-sectional regression.

8.8 Time Trends

Many economic time series exhibit persistent upward or downward trends.
A deterministic linear trend can be represented by including the time index t:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 t + u_t.$$

The coefficient β_2 captures the deterministic trend in y_t that is not explained by x_t.
Ignoring common trends can produce apparently strong relationships between variables even when the relationship is driven mainly by their shared movement over time.

8.9 Detrending

Including a time trend has an equivalent partialling-out interpretation.
Suppose

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_k x_{tk} + \gamma t + u_t.$$

First regress y_t on time:

$$y_t = a_0 + a_1 t + e_t.$$

Define the detrended dependent variable

$$\tilde{y}_t = y_t - \hat{a}_0 - \hat{a}_1 t.$$

For each explanatory variable, regress

$$x_{tj} = b_{0j} + b_{1j} t + v_{tj},$$

and define

$$\tilde{x}_{tj} = x_{tj} - \hat{b}_{0j} - \hat{b}_{1j} t.$$

Then regress

$$\tilde{y}_t = \beta_1 \tilde{x}_{t1} + \cdots + \beta_k \tilde{x}_{tk} + \tilde{u}_t.$$

By the partialling-out principle, the estimated slope coefficients are the same as those obtained from the original regression containing the time trend.
Thus including t effectively estimates the relationship between the detrended components of the variables.

8.10 Seasonality

Time series may also exhibit systematic seasonal patterns.
Monthly data may differ systematically across months, while quarterly data may differ across quarters.
Seasonality can be represented using dummy variables.
For quarterly data, define

$$Q_{2t}, \qquad Q_{3t}, \qquad Q_{4t},$$

with the first quarter as the base category.
A model with seasonal effects is

$$y_t = \beta_0 + \beta_1 x_t + \delta_2 Q_{2t} + \delta_3 Q_{3t} + \delta_4 Q_{4t} + u_t.$$

The coefficients $\delta_2, \delta_3, \delta_4$ measure systematic differences relative to the first quarter.

Example 8.10.1 (Seasonal Retail Sales)**.** Retail sales often rise substantially during particular months or quarters. A regression of sales on income that ignores seasonality may attribute part of this seasonal variation incorrectly to income.
Including seasonal dummy variables separates systematic calendar effects from the relationship of primary interest.

8.11 Stationary Stochastic Processes

Time series asymptotics require assumptions about how the probabilistic structure behaves over time.
Informally, a stationary process has a stable probabilistic structure.
A process $\{Y_t\}$ is strictly stationary if for every collection of time indices

$$t_1, \ldots, t_m$$

and every shift h, the joint distributions of

$$(Y_{t_1}, \ldots, Y_{t_m})$$

and

$$(Y_{t_1+h}, \ldots, Y_{t_m+h})$$

are identical.
A weaker and more commonly used condition is covariance stationarity.

Definition 8.11.1 (Covariance Stationarity)**.** A stochastic process $\{Y_t\}$ is covariance stationary if

$$\mathbb{E}(Y_t) = \mu$$

does not depend on t,

$$\mathrm{Var}(Y_t) = \sigma^2$$

does not depend on t, and

$$\mathrm{Cov}(Y_t, Y_{t+h})$$

depends only on the lag h, not on the calendar time t.

Thus the mean, variance, and autocovariance structure remain stable over time.

8.12 Weak Dependence

Stationarity alone does not require observations far apart in time to be nearly independent. For many large-sample results, an additional weak-dependence condition is imposed. Informally, a stationary process is weakly dependent if

$$\mathrm{Corr}(Y_t, Y_{t+h}) \to 0$$

as

$$h \to \infty.$$

Thus observations separated by a sufficiently large time interval have only weak dependence.
This property allows time-series versions of laws of large numbers and central limit theorems to operate.

8.13 Moving Average Processes

Let

$$\{e_t\}$$

be an i.i.d. sequence satisfying

$$\mathbb{E}(e_t) = 0, \qquad \mathrm{Var}(e_t) = \sigma_e^2.$$

Model 8.13.1 (MA(1) Process)**.** A moving average process of order one is

$$\boxed{X_t = e_t + \theta e_{t-1}}.$$

The current value depends on the current innovation and one lagged innovation.
The process is covariance stationary.
Moreover,

$$\mathrm{Cov}(X_t, X_{t-h}) = 0$$

for

$$h \geq 2.$$

Hence dependence disappears after one period, making the MA(1) process a simple example of a weakly dependent time series.

8.14 Autoregressive Processes

Another important class of models expresses the current value of a series in terms of its own past.

Model 8.14.1 (AR(1) Process)**.** An autoregressive process of order one is

$$\boxed{Y_t = \rho Y_{t-1} + e_t},$$

where $\{e_t\}$ is a zero-mean innovation process.

If

$$|\rho| < 1,$$

the process is stationary and weakly dependent.

Its autocorrelation function satisfies

$$\boxed{\text{Corr}(Y_t, Y_{t-h}) = \rho^h}.$$

Since

$$\rho^h \to 0$$

when $|\rho| < 1$, dependence decreases as the lag increases.
When ρ is close to 1, this decay is slow and the process is highly persistent.

8.15 Strict Exogeneity in Dynamic Models

The strict exogeneity condition

$$\mathbb{E}(u_t \mid X_1, \ldots, X_T) = 0$$

can be restrictive in dynamic models.
For example, consider

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t.$$

The lagged dependent variable y_{t-1} contains past shocks, so the relationship between regressors and errors must be examined carefully.
A weaker condition often considered in time-series analysis is contemporaneous exogeneity:

$$\mathbb{E}(u_t \mid X_t) = 0.$$

This condition is weaker than strict exogeneity because it does not require u_t to be unrelated to all future regressor values.
The exact condition required for consistency depends on the dynamic structure of the regression.

8.16 Random Walks

A central example of a nonstationary process is the random walk.

Model 8.16.1 (Random Walk)**.** A random walk is

$$\boxed{Y_t = Y_{t-1} + e_t},$$

where $\{e_t\}$ is a zero-mean innovation process.

Repeated substitution gives

$$Y_t = Y_0 + \sum_{j=1}^{t} e_j.$$

If Y_0 is fixed and

$$\text{Var}(e_t) = \sigma_e^2,$$

then

$$\mathbb{E}(Y_t) = Y_0$$

but

$$\boxed{\text{Var}(Y_t) = t\sigma_e^2}.$$

The variance therefore grows with time, so a random walk is not covariance stationary.
Shocks also have permanent effects. If e_t changes Y_t by one unit, that change remains embedded in all future values.

8.17 Random Walk with Drift

A random walk may contain a deterministic drift:

$$\boxed{Y_t = \alpha + Y_{t-1} + e_t}.$$

Repeated substitution yields

$$Y_t = Y_0 + \alpha t + \sum_{j=1}^{t} e_j.$$

Therefore,

$$\mathbb{E}(Y_t) = Y_0 + \alpha t,$$

while

$$\mathrm{Var}(Y_t) = t\sigma_e^2.$$

The process contains both a deterministic trend and a stochastic nonstationary component.

8.18 First Differences

For a random walk,

$$Y_t = Y_{t-1} + e_t.$$

Taking first differences gives

$$\Delta Y_t = Y_t - Y_{t-1} = e_t.$$

Thus although Y_t is nonstationary, its first difference may be stationary.
More generally,

$$\boxed{\Delta Y_t = Y_t - Y_{t-1}}$$

is the first-difference operator.
For a random walk with drift,

$$Y_t = \alpha + Y_{t-1} + e_t,$$

we obtain

$$\Delta Y_t = \alpha + e_t.$$

8.19 Order of Integration

The number of differences required to obtain a weakly dependent stationary process determines the order of integration.

Definition 8.19.1 (Integrated Process)**.** A stationary series is said to be integrated of order zero, written

$$Y_t \sim I(0).$$

If Y_t is nonstationary but

$$\Delta Y_t$$

is stationary, then

$$Y_t \sim I(1).$$

Thus a random walk is a standard example of an $I(1)$ process.
More generally, if d differences are required to obtain an $I(0)$ process, then

$$Y_t \sim I(d).$$

A highly persistent series may have

$$\text{Corr}(Y_t, Y_{t-1})$$

close to 1, which is an important warning sign of possible nonstationarity, although correlation alone is not a complete test for integration.

8.20 Trend-Stationary and Difference-Stationary Series

Two distinct sources of nonstationary-looking behavior should be distinguished.
A trend-stationary process has the form

$$Y_t = \alpha_0 + \alpha_1 t + v_t,$$

where v_t is stationary.
Removing the deterministic trend yields

$$Y_t - \alpha_0 - \alpha_1 t = v_t,$$

which is stationary.
By contrast, a difference-stationary process such as

$$Y_t = Y_{t-1} + e_t$$

contains a stochastic trend. Detrending alone does not remove the nonstationarity; first differencing is required:

$$\Delta Y_t = e_t.$$

This distinction matters because deterministic and stochastic trends require different transformations.

8.21 Using Trending Variables in Regression

If two unrelated variables both trend over time, a regression between them may produce a high R^2 and apparently significant coefficients simply because both series move persistently in the same direction.
For deterministic trends, one possible specification is

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 t + u_t.$$

For strongly persistent or integrated variables, first differences may be more appropriate:

$$\Delta y_t = \beta_0 + \beta_1 \Delta x_t + u_t.$$

The appropriate transformation depends on the time-series properties of the variables.

8.22 Large-Sample OLS with Time Series Data

For cross-sectional data, random sampling provides a convenient basis for laws of large numbers and central limit theorems.

Time series observations are generally not independent. Instead, stationarity and weak dependence can provide analogous large-sample conditions.

Under suitable assumptions,

$$\frac{1}{T}X^\top X \xrightarrow{p} Q$$

and

$$\frac{1}{T}X^\top \boldsymbol{u} \xrightarrow{p} 0.$$

Then

$$\boxed{\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}}.$$

Thus OLS consistency does not require complete independence across time, but sufficiently strong exogeneity and weak dependence are needed.

Under additional regularity conditions,

$$\sqrt{T}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$$

can also have an asymptotically normal distribution.

8.23 Summary of Time Series Regression

Time series regression differs from cross-sectional regression because the temporal ordering of observations carries statistical information.

Static models describe contemporaneous relationships, while distributed lag models allow the effect of an explanatory variable to unfold over time. Their coefficients distinguish immediate effects from the long-run propensity

$$\text{LRP} = \sum_{j=0}^{q} \delta_j.$$

Time trends and seasonal dummy variables capture deterministic temporal patterns. Detrending provides a partialling-out interpretation of regressions containing deterministic trends.

For stochastic processes, stationarity describes stability of the probability structure, while weak dependence requires observations far apart in time to become only weakly related. MA and stationary AR processes provide standard examples.

Random walks are fundamentally different:

$$Y_t = Y_{t-1} + e_t$$

is nonstationary, while

$$\Delta Y_t = e_t$$

may be stationary. This motivates the concept of integration and the distinction between $I(0)$ and $I(1)$ processes.

The remaining issue is dependence in the regression error itself. When

$$\text{Cov}(u_t, u_s \mid X) \neq 0$$

for different time periods, the regression contains serial correlation. Its consequences, tests, and corrections are developed in the next chapter.

Chapter 9

Endogeneity, Instrumental Variables, and 2SLS

9.1 The Endogeneity Problem

Consider the regression model

$$y = \beta_0 + \beta_1 x + u.$$

The central exogeneity condition for OLS is

$$\mathrm{Cov}(x, u) = 0.$$

If instead

$$\boxed{\mathrm{Cov}(x, u) \neq 0},$$

then x is endogenous.
Endogeneity may arise from omitted variables, measurement error, simultaneity, or other forms of dependence between the explanatory variable and the regression error.
Recall that in the simple regression model,

$$\operatorname{plim} \hat{\beta}_{1,OLS} = \beta_1 + \frac{\mathrm{Cov}(x, u)}{\mathrm{Var}(x)}.$$

Therefore, when

$$\mathrm{Cov}(x, u) \neq 0,$$

OLS is generally inconsistent.
Increasing the sample size does not solve the problem because the source of the error is systematic correlation between the regressor and the unobserved component.

9.2 Instrumental Variables

An instrumental variable provides an alternative source of variation in an endogenous explanatory variable.

Definition 9.2.1 (Instrumental Variable)**.** Suppose x is endogenous in

$$y = \beta_0 + \beta_1 x + u.$$

A variable z can serve as an instrument for x if it satisfies the following conditions.
First, z is excluded from the structural regression.
Second, z is correlated with the endogenous variable:

$$\boxed{\mathrm{Cov}(z, x) \neq 0}.$$

Third, z is uncorrelated with the structural error:

$$\boxed{\mathrm{Cov}(z, u) = 0}.$$

The second condition is called *instrument relevance*.
The third condition is called *instrument exogeneity* or *instrument validity*.
A useful instrument must satisfy both conditions.

9.3 Identification through an Instrument

Starting from

$$y = \beta_0 + \beta_1 x + u,$$

take the covariance with z:

$$\mathrm{Cov}(z, y) = \mathrm{Cov}(z, \beta_0 + \beta_1 x + u).$$

Since constants have zero covariance,

$$\mathrm{Cov}(z, y) = \beta_1 \,\mathrm{Cov}(z, x) + \mathrm{Cov}(z, u).$$

If z is exogenous,

$$\mathrm{Cov}(z, u) = 0,$$

so

$$\mathrm{Cov}(z, y) = \beta_1 \,\mathrm{Cov}(z, x).$$

If z is also relevant,

$$\mathrm{Cov}(z, x) \neq 0,$$

then

$$\boxed{\beta_1 = \frac{\mathrm{Cov}(z, y)}{\mathrm{Cov}(z, x)}}.$$

Thus the instrument identifies β_1 using the component of variation in x that is associated with z.

9.4 The Simple IV Estimator

The population covariance formula suggests the sample estimator

$$\boxed{\hat{\beta}_{1,IV} = \frac{\sum_{i=1}^{n}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{n}(z_i - \bar{z})(x_i - \bar{x})}}.$$

Equivalently,

$$\hat{\beta}_{1,IV} = \frac{\widehat{\mathrm{Cov}}(z, y)}{\widehat{\mathrm{Cov}}(z, x)}.$$

The intercept estimator is

$$\hat{\beta}_{0,IV} = \bar{y} - \hat{\beta}_{1,IV}\bar{x}.$$

The IV estimator does not use all variation in x. It uses the variation in x that can be explained by the instrument z.

9.5 Consistency of the IV Estimator

Using

$$y_i = \beta_0 + \beta_1 x_i + u_i,$$

the IV estimator satisfies

$$\begin{aligned}\hat{\beta}_{1,IV} &= \frac{\sum(z_i - \bar{z})(y_i - \bar{y})}{\sum(z_i - \bar{z})(x_i - \bar{x})} \\ &= \beta_1 + \frac{\sum(z_i - \bar{z})u_i}{\sum(z_i - \bar{z})(x_i - \bar{x})}.\end{aligned}$$

Dividing numerator and denominator by n,

$$\hat{\beta}_{1,IV} - \beta_1 = \frac{\frac{1}{n}\sum(z_i - \bar{z})u_i}{\frac{1}{n}\sum(z_i - \bar{z})(x_i - \bar{x})}.$$

Under suitable regularity conditions,

$$\frac{1}{n}\sum(z_i - \bar{z})u_i \xrightarrow{p} \text{Cov}(z, u)$$

and

$$\frac{1}{n}\sum(z_i - \bar{z})(x_i - \bar{x}) \xrightarrow{p} \text{Cov}(z, x).$$

Therefore,

$$\text{plim}\,\hat{\beta}_{1,IV} = \beta_1 + \frac{\text{Cov}(z, u)}{\text{Cov}(z, x)}.$$

If

$$\text{Cov}(z, u) = 0$$

and

$$\text{Cov}(z, x) \neq 0,$$

then

$$\boxed{\hat{\beta}_{1,IV} \xrightarrow{p} \beta_1}.$$

Thus IV can consistently estimate the coefficient even when

$$\text{Cov}(x, u) \neq 0.$$

9.6 The First-Stage Regression

Instrument relevance can be expressed through a regression of the endogenous variable on the instrument.
Consider

$$\boxed{x = \pi_0 + \pi_1 z + v}.$$

This is called the *first-stage regression.*
The relevance condition

$$\text{Cov}(z, x) \neq 0$$

corresponds to

$$\boxed{\pi_1 \neq 0}.$$

Thus instrument relevance can be investigated by testing

$$H_0 : \pi_1 = 0.$$

With one instrument, an ordinary t-test can be used.
If π_1 is close to zero, the instrument explains very little variation in x, and IV estimation may become highly imprecise.

9.7 Why IV Works

The first-stage regression decomposes x into

$$x = \hat{x} + \hat{v},$$

where

$$\hat{x} = \hat{\pi}_0 + \hat{\pi}_1 z.$$

Since $\hat{x}$ is constructed from the instrument, it contains the variation in x associated with z.
If

$$\text{Cov}(z, u) = 0,$$

this instrument-induced component of x is unrelated to the structural error.
IV therefore isolates a component of the endogenous variable that can be used to estimate the structural effect.

9.8 A Simple Economic Example

Example 9.8.1 (Endogenous Price)**.** Suppose the equation of interest is

$$Q = \beta_0 + \beta_1 P + u,$$

where Q is quantity demanded and P is price.
Price may be endogenous because market prices are jointly determined by supply and demand. Unobserved demand shocks contained in u can also affect the market price.
Hence

$$\text{Cov}(P, u) \neq 0,$$

and an OLS regression of quantity on price need not estimate the demand effect consistently.
An instrument z would need to affect price,

$$\text{Cov}(z, P) \neq 0,$$

while remaining unrelated to the demand error,

$$\text{Cov}(z, u) = 0.$$

The IV estimator then uses the variation in price generated by z, rather than all observed variation in price.

9.9 IV versus OLS

If x is genuinely exogenous, OLS is generally preferable because it uses all available variation in x.
IV uses only the part of the variation associated with the instrument, so its variance is typically larger.
Thus there is an important tradeoff:

OLS: more efficient if exogeneity holds

but

IV: consistent when the regressor is endogenous and the instrument is valid.

Therefore IV should not be introduced merely because an instrument is available. Its purpose is to address endogeneity.

9.10 Instrument Strength and Weak Instruments

Instrument relevance is not merely a binary condition.
Even if

$$\text{Cov}(z, x) \neq 0,$$

the correlation may be very small.
Then

$$\widehat{\text{Cov}}(z, x)$$

appears in the denominator of

$$\hat{\beta}_{1,IV} = \frac{\widehat{\text{Cov}}(z, y)}{\widehat{\text{Cov}}(z, x)},$$

so a small denominator can produce a highly variable estimator.
Such an instrument is called a *weak instrument*.
A weak first-stage relationship therefore leads to large IV standard errors and unstable coefficient estimates.
In practice, the first-stage regression should show that the excluded instruments provide meaningful explanatory power for the endogenous variable.

9.11 Structural Equations

Instrumental variables are particularly important in structural equations, where explanatory variables may be jointly determined with the dependent variable.
Consider

$$\boxed{y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1}.$$

Suppose y_2 is endogenous:

$$\text{Cov}(y_2, u_1) \neq 0.$$

The variable z_1 is exogenous and appears directly in the structural equation.
To estimate β_1, another exogenous variable z_2 may be used as an instrument for y_2, provided

$$\text{Cov}(z_2, u_1) = 0$$

and

$$\text{Cov}(z_2, y_2) \neq 0.$$

The variable z_2 is excluded from the structural equation but enters the reduced-form equation for y_2.

9.12 Reduced-Form Equation

Suppose

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1,$$

and z_2 is an excluded exogenous variable.
The endogenous variable can be modeled as

$$\boxed{y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + v_2}.$$

This is a reduced-form equation for y_2.
The relevance of z_2 requires

$$\boxed{\pi_2 \neq 0}.$$

Thus a t-test of

$$H_0 : \pi_2 = 0$$

provides a direct test of whether z_2 contributes to explaining the endogenous variable after controlling for the included exogenous variables.

9.13 Multiple Regression with Endogenous Variables

Consider a structural equation

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \beta_{p+1} w_1 + \cdots + \beta_k w_{k-p} + u,$$

where $x_1, \ldots, x_p$ are endogenous and $w_1, \ldots, w_{k-p}$ are exogenous.
Let

$$z_1, \ldots, z_m$$

denote additional excluded instruments.
The instrument set contains both

$$w_1, \ldots, w_{k-p}$$

and

$$z_1, \ldots, z_m.$$

The excluded instruments must satisfy the exogeneity conditions

$$\mathrm{Cov}(z_j, u) = 0$$

and must jointly explain the endogenous regressors.

9.14 Two-Stage Least Squares

When there are several instruments or several explanatory variables, instrumental-variable estimation is conveniently implemented using Two-Stage Least Squares.

Definition 9.14.1 (Two-Stage Least Squares)**.** Two-Stage Least Squares, abbreviated 2SLS, estimates a structural regression using predicted values of endogenous regressors obtained from the available exogenous variables and instruments.

Suppose the structural equation is

$$y = \beta_0 + \beta_1 x + \beta_2 w_1 + \cdots + \beta_k w_{k-1} + u,$$

where x is endogenous and the w_j's are exogenous.
Suppose $z_1, \ldots, z_m$ are excluded instruments.

9.15 Stage One of 2SLS

Regress the endogenous variable on all exogenous variables and all excluded instruments:

$$\boxed{x = \pi_0 + \pi_1 w_1 + \cdots + \pi_{k-1} w_{k-1} + \gamma_1 z_1 + \cdots + \gamma_m z_m + v}.$$

Obtain the fitted values

$$\hat{x}.$$

The fitted variable $\hat{x}$ is a linear combination of variables that are assumed to be exogenous. Thus the first stage extracts the component of x explained by the instrument set.

9.16 Stage Two of 2SLS

Replace the endogenous variable x by its fitted value $\hat{x}$ and estimate

$$\boxed{y = \beta_0 + \beta_1 \hat{x} + \beta_2 w_1 + \cdots + \beta_k w_{k-1} + \text{error}}.$$

The resulting coefficient estimates are the 2SLS estimates.
Conceptually,

$$\boxed{\text{Stage 1: isolate exogenous variation in } x}$$

and

$$\boxed{\text{Stage 2: use that variation to estimate the structural model.}}$$

9.17 2SLS with Multiple Instruments

Suppose the structural model is

$$y = \beta_0 + \beta_1 x + \beta_2 w + u$$

and two excluded instruments z_2 and z_3 are available.
Assume

$$\text{Cov}(z_2, u) = 0, \qquad \text{Cov}(z_3, u) = 0,$$

while both instruments are related to x.
The first-stage regression is

$$x = \pi_0 + \pi_1 w + \pi_2 z_2 + \pi_3 z_3 + v.$$

The fitted value is

$$\hat{x} = \hat{\pi}_0 + \hat{\pi}_1 w + \hat{\pi}_2 z_2 + \hat{\pi}_3 z_3.$$

Because $\hat{x}$ is a linear combination of the exogenous variables and valid instruments, it provides the instrumented component of x.
Instrument relevance requires at least one excluded instrument to contribute to the first stage:

$$\pi_2 \neq 0 \qquad \text{or} \qquad \pi_3 \neq 0.$$

A joint F-test can therefore be based on

$$H_0 : \pi_2 = \pi_3 = 0.$$

9.18 Identification

For IV estimation to be possible, the structural parameters must be identified.
A basic necessary condition is that the number of available excluded instruments be at least as large as the number of endogenous explanatory variables.
If there are p endogenous regressors and m excluded instruments, the order condition requires

$$\boxed{m \geq p}.$$

If

$$m < p,$$

the model is underidentified.
If

$$m = p,$$

the equation is exactly identified, provided the instruments satisfy the required rank condition.
If

$$m > p,$$

the equation is overidentified.
The order condition alone is not sufficient. The instruments must also contain enough independent variation to explain the endogenous regressors.
This requirement is summarized by the *rank condition*.

9.19 The Rank Condition

Suppose several endogenous variables are present.
The first-stage regressions must generate sufficiently independent variation in those endogenous variables.
It is not enough for every instrument to be individually correlated with some endogenous variable if all instruments effectively generate the same linear combination.
The relevant coefficient matrix in the reduced-form equations must have sufficient rank.
Thus identification requires both

$$\boxed{\text{sufficient number of instruments}}$$

and

$$\boxed{\text{sufficient independent first-stage variation.}}$$

9.20 Multicollinearity in 2SLS

Multicollinearity can also affect instrumental-variable estimation.
If the fitted endogenous variable $\hat{x}$ is strongly related to the other explanatory variables in the structural equation, the variance of the 2SLS estimator can become large.
More generally, weak or highly redundant instruments provide relatively little independent variation for identifying the endogenous regressor.
Thus 2SLS may have substantially larger standard errors than OLS.
This is another reason why IV estimation should be used to address a genuine endogeneity problem rather than automatically replacing OLS.

9.21 Testing for Endogeneity

Because OLS is generally more efficient when the explanatory variables are exogenous, it is useful to test whether endogeneity is present.
Suppose the structural model contains an explanatory variable x:

$$y = \beta_0 + \beta_1 x + \beta_2 w_1 + \cdots + \beta_k w_{k-1} + u.$$

First estimate the reduced-form equation

$$x = \pi_0 + \pi_1 w_1 + \cdots + \pi_{k-1} w_{k-1} + \gamma_1 z_1 + \cdots + \gamma_m z_m + v.$$

Obtain the residual

$$\hat{v}.$$

Then augment the structural equation:

$$y = \beta_0 + \beta_1 x + \beta_2 w_1 + \cdots + \beta_k w_{k-1} + \delta \hat{v} + e.$$

Test

$$\boxed{H_0 : \delta = 0}.$$

If H_0 is not rejected, there is no evidence from this test that x is endogenous.
If H_0 is rejected, the residual component of x is related to the structural error, providing evidence of endogeneity.
In that case, OLS is generally inconsistent and IV or 2SLS estimation is preferred when valid instruments are available.

9.22 OLS and 2SLS: Final Comparison

The difference between OLS and 2SLS can be summarized through the source of variation used for estimation.
OLS estimates the effect of x using all sample variation in x:

$$x = \text{exogenous variation} + \text{endogenous variation}.$$

If

$$\text{Cov}(x, u) = 0,$$

this is appropriate and OLS is typically more efficient.
If

$$\text{Cov}(x, u) \neq 0,$$

part of the variation in x is contaminated by the regression error.
IV and 2SLS instead use variation generated by valid instruments:

$$\boxed{\text{instrument} \longrightarrow \text{exogenous variation in } x \longrightarrow y.}$$

The price of removing endogenous variation is usually a loss of precision.
Thus the choice between OLS and IV is not simply

$$\text{small variance} \quad \text{versus} \quad \text{large variance}.$$

The fundamental distinction is

$$\boxed{\text{efficiency under exogeneity} \quad \text{versus} \quad \text{consistency under endogeneity}.}$$

9.23 Summary

The classical regression framework is built around the exogeneity condition

$$\text{Cov}(x, u) = 0.$$

When this condition fails, OLS is generally inconsistent.
An instrumental variable z addresses the problem by satisfying

$$\text{Cov}(z, x) \neq 0$$

and

$$\text{Cov}(z, u) = 0.$$

In the simple model,

$$\beta_1 = \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)},$$

which motivates the IV estimator.
With multiple explanatory variables or several instruments, the same principle leads to Two-Stage Least Squares:

first stage → predict endogenous regressors from instruments

followed by

second stage → estimate the structural equation using predicted values.

Instrument validity requires both relevance and exogeneity, while identification additionally requires sufficiently many independent instruments.
Compared with OLS, 2SLS generally has larger variance. Its purpose is not to improve an already valid OLS regression, but to obtain consistent estimation when endogeneity makes OLS unreliable.
Instrumental variables therefore complete the progression developed throughout these notes:

OLS → inference → asymptotics → violations of classical assumptions → alternative estimators.

Chapter 10

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